RHIN–VIOLA GROUPS ARE p-ADIC: THE TRANSFER PRINCIPLE, THE MARGIN MAP, AND THE OPTIMALITY OF THE KNOWN p-ADIC IRRATIONALITY MEASURES

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Abstract. Three p-adic irrationality measures are known: $\mu(\zeta_2(3)) \leq 7.177398\ldots$, $\mu(\zeta_3(3)) \leq 22.281447\ldots$ and $\mu(\zeta_2(5)) \leq 20.342651\ldots$. All three come from *totally symmetric* hypergeometric approximations, and Lai, Sprang and Zudilin close their note on $\zeta_2(5)$ by asking (i) whether the arithmetic group method of Rhin and Viola can be brought to bear on such constructions, and (ii) whether there is a hypergeometric-type identity for the Volkenborn integrals themselves. We answer both, and then map what the answers buy.

Our first result is a *Transfer Principle*. For $R \in \mathbb{Q}(t)$ with poles only at non-positive integers and $\deg R \leq -2$, the archimedean sum $\sum_{m\geq 0} R(m+\theta)$ and the Volkenborn integral $-\int_{\mathbb{Z}_p} \widetilde{R}(t+\theta)\,dt$ are linear forms with *the same* rational coefficients $\rho_{0,\theta}, \rho_2, \rho_3, \ldots$, produced by one and the same $\mathbb{Q}$-linear functional of the partial-fraction data of R. Hence every Rhin–Viola/Zudilin transfer identity, and every denominator lemma, is simultaneously an archimedean and a p-adic statement: question (ii) needs no new Volkenborn transformation theory, because the group acts on the rational function and the Volkenborn integral is a passive spectator. Concretely, we exhibit the order-1920 group $W(D_5)$ acting on the weight-3 family, verify its transfer identity coefficientwise at 1200 orbit points with 0 failures — and, decisively for the p-adic application, at 780 points with half-integer parameters — and validate the resulting savings function end to end by recovering Rhin and Viola's $\mu(\zeta(3)) \leq 5.51389062$ as 5.51389063. A sweep of a 37-paper corpus finds no prior p-adic use of a Rhin–Viola group.

The rest of the paper answers question (i) quantitatively, and the answer is negative in a sharp and informative way. We set up a parametrised Volkenborn construction and an exact cost ledger — growth G, smallness α, denominator cost E, margin $= \alpha - G - E$ — whose only inputs are the parameter tuple $(p, r, \text{shifts}, A, m)$ and the well-poised flag. Fed the published parameters, it returns all three known measures to every printed digit, together with four further independent anchors. We prove that a two-term p-adic form in 1 and $\zeta_p(w)$ arises from a *single integration shift* exactly when $\varphi(p^r) = 2$, i.e. $p^r \in \{3, 4\}$ (plus the exceptional point $(p, r) = (2, 1)$); that for $p \geq 5$, whenever the coset count exceeds the number of distinct brick shifts, the p-adic smallness of the twisted coset sum cancels the archimedean coefficient growth exactly, leaving margin $= -E$; and, for the very-well-poised family at $p = 2$, a parity law that collapses the whole ledger to $\text{margin}(M, m) = 2M \log 2 - (M + m)$.

Three computations then close the space. First, over the cone searched — $O(1)$ insets, scaled insets, all rank-reductions by polynomial-coefficient operators, and the group orbit — the three published measures are *exactly optimal*, with no improvement anywhere; the group saving $\delta_{\mathfrak{G}}$ is provably 0 at the symmetric point and is nowhere positive on the part of the p-adic locus we search. Second, the rank cannot be cut: five elimination searches over inset cones return 0/300 surviving operators against positive controls at 300/300. Third, the one escape hatch at $p \geq 5$ — cancellation between cosets — is shut over a fair sweep: gains over the per-coset minimum are ≤ 3 *absolutely*, and the hyperplane depth of the coset vector is bounded and constant in n, while the requirement grows like $E \cdot n$. The resulting margin map exhibits a clean dichotomy: *at $p \in \{2, 3\}$ the binding wall is rank; at $p \geq 5$ it is size*, and after a correction to E (a second prime band on $(n, 2n)$, invisible to every published anchor) the nearest misses in the whole family are $\zeta_5(3)$ and $\zeta_7(3)$ at exactly -3.000.

The structural finding is a disjoint-support tension: the archimedean coefficient growth C_1, which Bel's p-adic criterion must pay and the archimedean Rhin–Viola criterion never does, vanishes exactly on the degenerate locus where the group saving vanishes. Buying $\delta_{\mathfrak{G}} > 0$ switches C_1 on, and C_1 alone exceeds the entire available $\delta_{\mathfrak{G}}$. We isolate two open lemmas that would upgrade the two obstructions to theorems, and describe what a construction evading the map would have to look like. Optimality claims are scoped throughout to the stated family and cone; evidence classes are attached to every assertion, and a reproduction appendix lists the programs.

Contents

1. Introduction

1.1. **The three known measures.** Let p be a prime and let ζ_p denote the Kubota–Leopoldt p-adic zeta function, normalised as in [14] so that $\zeta_p(s) = L_p(s, \omega^{1-s})$ for integers $s \geq 2$. Only three p-adic irrationality *measures* are on record. Writing $\mu(\xi)$ for the irrationality exponent of $\xi \in \mathbb{Q}_p \setminus \mathbb{Q}$ in the p-adic sense of Bel [3], they are

$$\begin{aligned}
\mu(\zeta_2(3)) &\leq \frac{12\log 2}{6\log 2 - 3} = 7.177398\ldots,\\
\mu(\zeta_3(3)) &\leq \frac{6\log 3}{3\log 3 - 3} = 22.281447\ldots, \qquad (1.1)\\
\mu(\zeta_2(5)) &\leq \frac{16\log 2}{8\log 2 - 5} = 20.342651\ldots.
\end{aligned}$$

The first two are extracted in [14, §1] from Calegari's work [8] by way of Bel's lemma, or equivalently from Beukers [4, Thm. 11.2]; the third is the main theorem of [14]. The corresponding irrationality statements go back to Calegari [8] and Beukers [4] for weight 3, and to [14] for $\zeta_2(5)$; the many-values theorems of Lai–Sprang [13] and Lai [11] produce irrationality of *one of* a list, never a measure for an individual value.

Every one of the constructions behind (1.1) is *totally symmetric*: the underlying rational function has all its numerator bricks of the same length and all its pole bricks of the same length, sitting on the same interval. In the archimedean world that is precisely the configuration on which the Rhin–Viola arithmetic group method [16], and its hypergeometric reformulation by Zudilin [17], buy *nothing*: the orbit of a totally symmetric point is a single point (Proposition 3.8). The archimedean record $\mu(\zeta(3)) \le 5.51389062$ of Rhin and Viola lives instead at a strongly asymmetric point, where the group removes about 40% of the denominator (§3.5).

1.2. **The questions we answer.** Lai, Sprang and Zudilin end [14] with two explicit invitations. The first [14, Final remarks]:

> "Speaking about the irrationality measure for $\zeta_2(5)$ and also for the related p-adic zeta values $\zeta_2(3), \zeta_3(3)$, our construction in this note and the earlier ones in [Beukers, Calegari, Lai] exploit exclusively the so-called totally symmetric hypergeometric approximations (or their equivalents). It may certainly be of interest to take a step further and investigate more general hypergeometric series leading to rational approximations that depend not only on $n \in \mathbb{Z}_{\ge 0}$ but on multiple integer parameters. Such generalisations are amenable to additional arithmetic manipulations, for example, to the use of the arithmetic group method as developed in the works of Rhin and Viola."

The second, after observing that Lai's and Beukers' two rational functions produce the *same* linear forms in 1 and $\zeta_2(3)$ [14, Final remarks]:

> "It is natural to expect a hypergeometric-type identity for the Volkenborn integrals behind the coincidence $S_n^{(\mathrm{L})} = S_n^{(\mathrm{B})}$."

Independently, Brown [6, Rem. 47] records a referee's question in the same spirit for his determinant criterion: "A referee asked the excellent question of whether the matrices D_n can be designed to make use of the cancellation of primes in the method of Rhin–Viola. I do not know the answer to this question, but the numerical experiments to be discussed later clearly demonstrate a large degree of 'prime cancellation' of possibly a different nature."

This paper answers the second question completely and the first question quantitatively — the latter in the negative, but in a way that we believe is more useful than a positive answer of the expected size would have been, because it comes with a map.

1.3. **Results. (A) The Transfer Principle** (Theorem 1, [PROVED]). Let $R \in \mathbb{Q}(t)$ have poles only at non-positive integers and $\deg R \le -2$, with partial fractions $R(t) = \sum_{i,k} r_{i,k}/(t+k)^i$, and set

$$\rho_i := \sum_k r_{i,k}, \qquad \rho_{0,\theta} := -\sum_{i,k} r_{i,k} \sum_{\nu=0}^{k-1} (\nu+\theta)^{-i}.$$

Then, with $\widetilde{R}$ a primitive of R,

$$\sum_{m \ge 0} R(m+\theta) = \rho_{0,\theta} + \sum_{i \ge 2} \rho_i \, \zeta(i,\theta), \qquad -\int_{\mathbb{Z}_p} \widetilde{R}(t+\theta)\,\mathrm{d}t = \rho_{0,\theta} + \sum_{i \ge 2} \rho_i \, \omega(\theta)^{1-i} \zeta_p(i,\theta),$$

with the same rational numbers $\rho_{0,\theta}, \rho_2, \rho_3, \ldots$. The two constructions differ only in which *value* is attached to each ρ_i; the coefficient map $R \mapsto (\rho_{0,\theta}, \rho_2, \rho_3, \ldots)$ is literally the same $\mathbb{Q}$-linear functional of the partial-fraction data. The immediate corollary (Corollary 2.5) is that every Rhin–Viola/Zudilin transfer identity, being an identity between the rational coefficients at two parameter points, is simultaneously an archimedean and a p-adic identity; likewise every denominator and integrality lemma. In particular the second LSZ question is answered by observing that no hypergeometric-type identity *for the Volkenborn integrals* is needed: the identity lives on the rational coefficients, and the Volkenborn integral never sees it.

(B) The group, made explicit and made p-adic (§3). For the weight-3 family we realise Zudilin's group $\mathfrak{G} = \langle \mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3, \mathfrak{b}, \mathfrak{h}\rangle$ of order $1920 = |W(D_5)|$ [VERIFIED: BFS closure, exact], exhibit its invariant in the h-parametrisation, and verify the transfer identity coefficientwise — on both the rational part and the $\zeta(3)$-part — at every point of the orbits of four directions, 1200/1200 at the generic direction with 0 failures [VERIFIED: 4 directions, orbits of size $1, 40, 80, 1920$]. Crucially for the p-adic application, the same test passes at *half-integer* parameters, 780/780 comparable points with 0 failures: the mechanism is lattice-robust, not an accident of integrality. The savings function $\delta_{\mathfrak{G}}$ is implemented and validated end to end by recovering Rhin and Viola's published $\mu(\zeta(3)) \le 5.51389062$ as 5.51389063 [VERIFIED: 8 digits]. A cross-tabulated sweep of a 37-paper corpus for p-adic markers against group markers returns no paper using a Rhin–Viola group p-adically [VERIFIED: corpus of 37 papers, §3.8].

(C) The ledger and its validation (§4, §5). We write down a parametrised Volkenborn construction containing, as exact special cases with their printed constants, the constructions of Beukers, Calegari (through Beukers' reformulation), Lai, Lai–Sprang, Lai–Lupu–Sprang and LSZ, and derive an exact cost ledger

$$G = v_p(C)\log p, \qquad \alpha = \Big[v_p(C) + Ar + \min_{\theta_0\in\Theta}\sum_c \mathrm{ctb}_c(\theta_0)\Big]\log p,$$
$$\beta = G + E, \qquad \text{margin} = \alpha - \beta,$$

in which every component is separately derived and separately measured. Fed *only* the parameter tuple, the ledger returns all three measures (1.1) to every printed digit, plus four further independent anchors. This validation section is the load-bearing part of the paper's credibility: nothing downstream is calibrated.

(D) The classification (Theorem 2, [PROVED]). A single Volkenborn form in 1 and the *full* value $\zeta_p(w)$, rather than in Hurwitz values $\zeta_p(w,\theta)$, exists exactly when $\varphi(p^r) = 2$, i.e. $p^r \in \{3,4\}$, plus the exceptional point $(p,r) = (2,1)$ where the single shift $\frac{1}{2}$ already *is* the full ζ_2. Otherwise the full twisted coset sum is forced, and the p-adic smallness becomes a minimum over $\varphi(p^r)$ cosets. At $p \ge 5$, whenever $\varphi(p^r)$ exceeds the number of distinct numerator-brick shifts, that minimum is attained at a coset with no aligned brick, and then $Ar + \sum_c \mathrm{ctb}_c = 0$ *exactly*: the p-adic smallness cancels the archimedean growth to the last digit and the entire d_n^E denominator cost goes unpaid, so margin $= -E$ (Proposition 6.1). Without that hypothesis the deficit is bounded rather than equal to $-E$ (Corollary 6.4, Remark 6.2).

(E) Optimality of the three known measures (Theorem 3, [COMP: stated cone]). Over the cone described precisely in §7.2 — all $O(1)$ insets of the bricks, all scaled (Rhin–Viola) insets, all rank reductions by polynomial-coefficient difference operators, and the whole 1920-element group orbit — the bounds (1.1) are attained and cannot be improved:

$$\mu(\zeta_2(3)) \le 7.17739889912418, \quad \mu(\zeta_3(3)) \le 22.28144795149432, \quad \mu(\zeta_2(5)) \le 20.34265173891448,$$

with E and the archimedean growth *measured* rather than modelled, so that any Φ_n-type or Rhin–Viola saving actually present is already counted. As stressed in §1.4, this is optimality over a family, not an absolute statement.

(F) The rank wall (§7.5). For the very-well-poised $p=2$ family a parity law forces $\rho_i = 0$ whenever $i+M$ is even (Proposition 7.3), whence the whole ledger collapses to $\mathrm{margin}(M,m) = 2M\log 2 - (M+m)$, which tends to $+\infty$: *size never binds at* $p=2$. The sole obstruction is that the form carries $\lfloor (M-1)/2\rfloor$ zeta values. Removing a coefficient requires a difference operator with polynomial coefficients; five searches over inset cones (dimensions $144, 210, 144, 144, 112$) return 0 out of 300 kernel vectors that keep the wanted coefficient, against positive controls at 300/300 [VERIFIED: $\mathbb{F}_q$, $q = 2^{61}-1$, every hit re-verified over $\mathbb{Q}$].

(G) The shut hatch (§7.8). The only place where the $p \geq 5$ verdict could be overturned inside this parameter space is cancellation between cosets, since $\alpha = \min_j$ is an inequality. Exact computation of every $S_n(j/p^r)$ shows the plain and all ω-twisted sums beat the per-coset minimum by at most 3 *absolutely*, not per n; and the Smith-normal-form hyperplane depth of the coset vectors is bounded by 28 and constant in n, while the requirement grows like $E \cdot n$ [VERIFIED: $p \in \{5, 7\}$, $r \in \{1, 2\}$, $A \in \{2, 3, 4\}$, $m \in \{0, 1\}$, $n \leq 30$].

(H) The map and the dichotomy (§8). Putting the above together gives a complete margin table over $p \leq 13$, $w \leq 9$ and rank ≤ 4, in two versions: with the ledger's E, and with the corrected, measured E_{true}. The headline is a dichotomy that the campaign literature has been treating as one obstruction:

at $p \in \{2, 3\}$ the binding wall is rank; at $p \geq 5$ it is size.

After the E-correction the nearest misses in the whole family are $\zeta_5(3)$ and $\zeta_7(3)$ at exactly -3.000, and those are structurally the *hardest* cells, since there the deficit is $-E$ exactly.

(I) The structural finding (§9.3). Bel's p-adic criterion requires $\alpha >$ growth + denominators, whereas the archimedean Rhin–Viola criterion requires only $C_0 > C_2$: the archimedean coefficient growth C_1 enters the archimedean *measure* but never the archimedean *irrationality condition*, while p-adically it must be paid. And $C_1 = 0$ holds exactly on the degenerate locus where the numerator bricks overlie the poles — which is exactly where the group saving $\delta_{\mathfrak{G}}$ vanishes. $\delta_{\mathfrak{G}} > 0$ *and* $C_1 = 0$ *have disjoint support.* Measured, $C_1/$ budget is 1.175, 0.969, 1.073 at three test directions, each alone exceeding the entire available $\delta_{\mathfrak{G}}/$ budget ≤ 0.44.

1.4. **Scope of the optimality claims.** We state this once, prominently, and repeat it at each use. *No claim in this paper is that the bounds* (1.1) *are unimprovable.* What we prove and compute is optimality *over the family and cone that we search*, which is stated explicitly, with the search domain written out, at every occurrence. The family is the parametrised Volkenborn construction of Definition 4.1 together with the inset cones of §7.2 and the difference-operator module of §7.5. A construction outside that family is not covered; §9.4 describes what such a construction would have to look like, precisely so that the boundary of our claim is visible.

Similarly, the negative results of §7.5 and §7.8 are fair sweeps with clean negatives and stated ranges, backed by positive controls; they are not impossibility proofs. §9.1 isolates the two lemmas that would convert them into theorems.

1.5. **Evidence classes.** Every substantive assertion below carries one of the following tags. This convention is inherited from the working memos underlying this paper and is used without exception.

- [PROVED] — a proof is given here, or read out of a proof in a cited source, with the source lemma named.
- [DERIVED] — derived here from cited lemmas by a chain that is written out, but not separately measured.
- [VERIFIED: range] — an exact finite computation in exact rational or integer arithmetic; the range of parameters and the sample size are always stated inside the tag.
- [COMP: domain] — a computed optimum, exhaustive over the stated domain; the domain is always stated inside the tag.
- [EXCLUDED: tolerance, bounds] — a search that returned nothing, with the tolerance and the coefficient/degree bounds of the search stated.

- [CALIBRATED] — a modelling assumption. Exactly one such statement appears in this paper (§9.2), it is flagged in place, and it was subsequently *refuted* by the first-principles computation of §9.3; we retain it because the refutation is the structural finding.
- [OPEN] — a statement we believe and do not prove.

All arithmetic reported as exact is carried out in `fractions.Fraction` or in exact integer arithmetic; p-adic valuations come from exact truncated Volkenborn series. Wide combinatorial searches are run over $\mathbb{F}_q$ with $q = 2^{61} - 1$ and every hit is re-verified over $\mathbb{Q}$. Floating point appears only in the final evaluation of transcendental constants such as $\log 2$, and the headline constants are re-verified at 50 decimal digits.

Remark 1.1 (Methodology; flagged)**.** The computations reported here were carried out with substantial machine assistance, including automated search over parameter cones and over candidate difference operators; every positive identification was re-derived independently of the search that proposed it, every negative is reported with its search domain, tolerance and positive control, and the programs are listed in Appendix A.

1.6. **Notation and one renaming.** p is a prime, $q_p = p$ for odd p and $q_2 = 4$, φ is Euler's totient, v_p is the p-adic valuation with $v_p(p) = 1$, and $d_n = \operatorname{lcm}(1, \ldots, n)$. We write ω for the Teichmüller character $\mathbb{Z}_p^\times \to \mu_{\varphi(q_p)}(\mathbb{Z}_p)$, *extended to* $\mathbb{Q}_p^\times$ as in [13, §2] by

$$\omega(x) := p^{v_p(x)}\,\omega\big(x/p^{v_p(x)}\big), \qquad \langle x\rangle := x/\omega(x). \tag{1.2}$$

The extension is what carries the powers of p in (2.5) and (4.2). We write $(x)_L = x(x+1)\cdots(x+L-1)$ for the Pochhammer symbol and $\zeta(s,\theta)$, $\zeta_p(s,\theta)$ for the Hurwitz and p-adic Hurwitz zeta functions. Following [13, Lemma 12] we use

$$D^i\,\zeta_p(i) = \sum_{\substack{1\le j\le D\\ \gcd(j,p)=1}} \omega\Big(\frac{j}{D}\Big)^{1-i}\zeta_p\Big(i, \frac{j}{D}\Big), \qquad q_p \mid D,\ i \ge 2. \tag{1.3}$$

Convention 1.2 (the well-poised flag)**.** In the source memos the very-well-poised indicator of the rational function is written $\delta \in \{0,1\}$, and the Rhin–Viola denominator saving is *also* written δ. Since both appear in the same formulas here, we write $\epsilon \in \{0,1\}$ for the well-poised flag (the exponent of the factor $2t + h_0$) and reserve $\delta_{\mathfrak{G}} = \lim_n \log \Phi_n/n$ for the group saving. Consequently the denominator cost reads $E = A + m + 1 - \epsilon$ and never $A + m + 1 - \delta$.

1.7. **Plan.** §2 proves the Transfer Principle and its corollaries. §3 makes the group explicit, records the verification of the transfer identity at integer and at half-integer parameters, states the four honest flags (F1)–(F4) including the Gauss-duplication bridge, and reports the corpus sweep. §4 sets up the parametrised construction and the exact ledger. §5 is the validation section. §6 proves the $\varphi(p^r) = 2$ classification and the $p \ge 5$ exact cancellation. §7 reports the scan: optimality, the rank wall, the shut hatch. §8 gives the margin map and the dichotomy. §9 is the discussion: the two open lemmas, the disjoint-support tension, and what a construction evading the map would need. Appendix A is the reproduction record.

2. The Transfer Principle

This section contains the load-bearing new statement of the paper. It is elementary; its content is entirely in what it says about the relation between two constructions that the literature has treated as incompatible.

2.1. **The coefficient functional.**

Definition 2.1. Let $R \in \mathbb{Q}(t)$ have poles only at non-positive integers and $\deg R \le -2$, and write its partial-fraction decomposition as

$$R(t) = \sum_{i\ge1}\sum_{k\ge0} \frac{r_{i,k}}{(t+k)^i}, \qquad r_{i,k} \in \mathbb{Q}, \tag{2.1}$$

the sums being finite. For θ in the domain of definition below, set

$$\rho_i := \sum_k r_{i,k} \quad (i \ge 1), \qquad \rho_{0,\theta} := -\sum_{i,k} r_{i,k} \sum_{\nu=0}^{k-1} (\nu+\theta)^{-i}, \tag{2.2}$$

with the convention that the empty sum $\sum_{\nu=0}^{-1}$ is 0. We call

$$\mathcal{R}: \ R \longmapsto (\rho_{0,\theta}, \rho_2, \rho_3, \rho_4, \dots) \tag{2.3}$$

the *coefficient functional* of R at θ. It is manifestly $\mathbb{Q}$-linear in the partial-fraction data $(r_{i,k})$, and for fixed $\theta \in \mathbb{Q}$ it takes values in $\mathbb{Q}$.

We note at once the vanishing that both worlds use.

Lemma 2.2 (purity by degree)**.** *[PROVED] If* $\deg R \le -2$ *then* $\rho_1 = \lim_{t\to\infty} tR(t) = 0$.

Proof. Multiplying (2.1) by t and letting $t \to \infty$, every term with $i \ge 2$ contributes 0 and the terms with $i = 1$ contribute $\sum_k r_{i,k} = \rho_1$; on the other hand $tR(t) \to 0$ because $\deg R \le -2$. ☐

2.2. **The principle.**

Theorem 1 (Transfer Principle)**.** *[PROVED] Let R be as in Definition 2.1 and let $\widetilde{R}$ be a primitive of R (any antiderivative; the two statements below are insensitive to the constant, which is killed by $\rho_1 = 0$).*

(i) (archimedean) *For $\theta > 0$,*

$$\sum_{m\ge0} R(m+\theta) \ = \ \rho_{0,\theta} + \sum_{i\ge2} \rho_i\, \zeta(i,\theta). \tag{2.4}$$

(ii) (p-adic) *For $\theta \in \mathbb{Q}_p$ with $|\theta|_p \ge q_p$ — the domain on which $\zeta_p(s,\theta)$ is defined* [13, §2] —

$$-\int_{\mathbb{Z}_p} \widetilde{R}(t+\theta)\,\mathrm{d}t \ = \ \rho_{0,\theta} + \sum_{i\ge2} \rho_i\, \omega(\theta)^{1-i} \zeta_p(i,\theta). \tag{2.5}$$

The rational numbers $\rho_{0,\theta}$ and ρ_i occurring in (2.4) *and in* (2.5) *are* the same*: both are the values of the single functional* (2.3).

Remark 2.3 (the excluded point $\theta = \frac12$, $p = 2$)**.** Since $q_2 = 4$ and $|\frac12|_2 = 2$, the hypothesis of (ii) excludes $\theta = \frac12$ at $p = 2$ — which is exactly the shift used by every $p = 2$ construction below. There $\zeta_2(s, \frac12)$ is not defined, and its role is played instead by [14, Lemma 9],

$$\frac{1}{s-1} \int_{\mathbb{Z}_2} \frac{\mathrm{d}t}{(t+\frac12)^{s-1}} = 2^s\, \zeta_2(s) \qquad (s \in \mathbb{Z}_{\ge2}),$$

in which the single shift $\frac12$ already produces the full ζ_2. Statement (i), the coefficient functional (2.3) and both corollaries of §2.3 are unaffected, because they are statements about $\mathbb{Q}$; what changes at $(p,r) = (2,1)$ is only which p-adic *value* is attached to ρ_i. Everywhere else in the paper $|\theta_0|_p \ge q_p$ holds.

Proof. Statement (ii) is [13, Lemma 21], whose proof we recall in outline because it is what makes the comparison visible. Writing $\widetilde{R}(t+\theta) = \sum_k r_{1,k} \log_p\langle t+k+\theta\rangle + \sum_{i\geq 2}\sum_k r_{i,k}(1-i)^{-1}(t+k+\theta)^{1-i}$, one integrates termwise; the translation formula for the Volkenborn integral [14, Lemma 5] moves each shift k back to 0 at the cost of the finite sum $\sum_{\nu<k}(\nu+\theta)^{-i}$, and the identity $\zeta_p(i,\theta) = \omega(\theta)^{i-1}\cdot\frac{1}{i-1}\int_{\mathbb{Z}_p} \mathrm{d}t/(t+\theta)^{i-1}$ of [13, §2] converts the remaining integrals. The $\log_p$ terms carry the factor ρ_1, which vanishes by Lemma 2.2. Collecting the finite sums gives exactly $\rho_{0,\theta}$ of (2.2).

Statement (i) is the classical Hurwitz expansion, obtained by the identical bookkeeping. Summing (2.1) over $m \geq 0$ termwise, $\sum_{m\geq 0}(m+k+\theta)^{-i} = \zeta(i,\theta) - \sum_{\nu=0}^{k-1}(\nu+\theta)^{-i}$ for $i \geq 2$, which produces $\sum_i \rho_i\zeta(i,\theta)$ together with precisely the finite correction $\rho_{0,\theta}$; the $i=1$ terms again assemble into ρ_1 times a divergent constant and vanish by Lemma 2.2, absolute convergence of the whole rearrangement being guaranteed by $\deg R \leq -2$.

The final sentence is not an additional claim but an observation about the two computations just performed: in each, the coefficient of $\zeta(i,\cdot)$ resp. $\zeta_p(i,\cdot)$ is $\sum_k r_{i,k}$, and the rational term is $-\sum_{i,k} r_{i,k}\sum_{\nu<k}(\nu+\theta)^{-i}$. Neither expression refers to the completion. □

Remark 2.4. The two constructions differ *only* in which value is attached to each ρ_i: $\zeta(i,\theta)$ on one side, $\omega(\theta)^{1-i}\zeta_p(i,\theta)$ on the other. This is the exact sense in which they are "the same construction". It is also, we note, the precise content of the correspondence that LSZ point out in [14, Remark 13], where the coefficients $\rho_{n,0}, \rho_{n,3}$ of their p-adic form are observed to be "exactly the same" as those of the archimedean expansion $\sum_{m\geq 0} R_n''(m+\frac{1}{2}) = \rho_{n,0} + \rho_{n,3}(1-2^{-5})\zeta(5)$. What is new here is not the observation at one point but its use as a *principle.*

2.3. **The two corollaries that do the work.**

Corollary 2.5 (transfer of identities)**.** *[PROVED] Let $\mathcal{F}$ be any family of rational functions as in Definition 2.1, parametrised by a set P, and let $g : P \to P$ be a map for which one knows an identity of the form*

$$\mathcal{R}(R_{g\mathbf{a}}) = \kappa(\mathbf{a})\cdot\mathcal{R}(R_{\mathbf{a}}), \qquad \kappa(\mathbf{a}) \in \mathbb{Q}^\times,$$

i.e. a coefficientwise proportionality of the two coefficient vectors. Then that identity holds verbatim for the archimedean linear forms (2.4) *and for the Volkenborn linear forms* (2.5) *at every admissible θ, with the same κ.*

Proof. Immediate from Theorem 1: both forms are the images of the same vector $\mathcal{R}(R)$ under a fixed (completion-dependent) evaluation map, and the hypothesis is a statement about $\mathcal{R}(R)$ alone. □

Corollary 2.6 (transfer of denominators)**.** *[PROVED] Any statement of the form "$\Lambda\cdot\rho_i \in \mathbb{Z}$ for all i" or "$\Lambda\cdot\rho_{0,\theta} \in \mathbb{Z}$", with $\Lambda \in \mathbb{Q}$ depending on the parameters only, is a statement about rational numbers and therefore holds simultaneously for the archimedean and the p-adic construction built from the same R.*

Corollaries 2.5 and 2.6 are the answer to the second LSZ question of §1.2. The expected shape of an answer was a summation or transformation theory for Volkenborn integrals. The actual answer is that none is required: the group acts on R, the transformation identity is an identity between the rational vectors $\mathcal{R}(R)$, and the Volkenborn integral is a passive spectator. In the terminology of the campaign this is the p-adic analogue of the ν_p-sharpening principle, pushed one level further — not merely "valid in every completion" but "the p-adic construction's coefficients *are* the archimedean construction's coefficients".

Remark 2.7. Three things Theorem 1 does *not* give, stated here so that the reader is not misled by §3. (a) It does not say that the p-adic *smallness* transfers; smallness is a statement about the value, not the coefficients, and it is exactly where the two worlds diverge (§4). (b) It does not say that admissibility transfers; a rational function whose numerator bricks sit at the wrong shift is a perfectly good archimedean object and a bad p-adic one (Proposition 3.11). (c) It does not give a group where there is none: the family LSZ actually use has trivial transfer structure (Finding 3.12).

2.4. **The savings function is completion-independent.** For the Rhin–Viola machinery one needs to know that the denominator saving is the same object in both worlds. It is, and for a cheap reason.

Proposition 2.8. *[DERIVED] Let the parameters of a hypergeometric brick be integers $c \cdot n$ with c a fixed direction vector, and let $\Phi_n = \prod_{\sqrt{m_0 n}<\ell\le m_3 n} \ell^{\nu_\ell}$ be the Rhin–Viola/Zudilin denominator saving, $\nu_\ell = \max_{g\in\mathfrak{G}} \operatorname{ord}_\ell(\Pi(cn)/\Pi(gcn))$. Then, replacing each integer parameter $c\cdot n$ by a θ-shifted parameter $c\cdot n + O(1)$ leaves*

$$\delta_{\mathfrak{G}} := \lim_{n\to\infty} \frac{\log \Phi_n}{n}$$

unchanged, so that $\delta_{\mathfrak{G}}$ is the same function of the direction in the p-adic and in the archimedean setting.

Argument. An $O(1)$ shift changes ord_ℓ of each factorial by at most $O(1)$ and hence each ν_ℓ by $O(1)$. Bounding each of the $\pi(m_3 n) - \pi(\sqrt{m_0 n}) = O(n/\log n)$ primes in the window by its own $O(\log \ell) = O(\log n)$ gives only $O(n)$, which is not enough; the point is that the shift moves $\lfloor c_i n/\ell \rfloor$ only when a carry occurs, i.e. for a set of primes of relative density $O(1/\ell)$, so that the expected total displacement is $O\big(\sum_{\ell\le m_3 n} \log \ell/\ell\big) = O(\log n) = o(n)$. Equivalently and more structurally: with the shift absorbed into the direction, the perturbed direction is $c + O(1/n) \to c$, and $\delta_{\mathfrak{G}}$ is given by the integrals (3.7) of the piecewise constant Λ, whose discontinuity set is finite and $\mathrm{d}\psi$-null away from 0; the integrals are therefore continuous along $c + O(1/n) \to c$. We do not carry the second argument out in detail, and label the statement [DERIVED] accordingly. $\square$

Proposition 2.8 is what licenses computing $\delta_{\mathfrak{G}}$ entirely with the classical archimedean machinery, which is what §3 does.

3. The group, and what it looks like p-adically

3.1. **The group, explicitly.** We use Zudilin's realisation [17, Lemma 8] of the Rhin–Viola group for weight 3. Directions are pairs $(\alpha, \beta) = (\alpha_1, \dots, \alpha_4; \beta_1, \dots, \beta_4)$ of integer vectors with

$$\{\beta_1, \beta_2\} < \{\alpha_1, \dots, \alpha_4\} < \{\beta_3, \beta_4\}, \qquad \textstyle\sum_j \alpha_j = \sum_k \beta_k, \tag{3.1}$$

and the group acts by permutations of the sixteen entries of the matrix

$$c_{jk} = |\alpha_j - \beta_k|, \qquad j, k = 1, \dots, 4. \tag{3.2}$$

Zudilin's Lemma 8 writes $c_{jk} = b_k - a_j - 1$ in the case $a_j < b_k$; under (3.1) that differs from (3.2) by 1 in the two entries c_{33}, c_{44}. Since the entries are scaled by n, the two conventions differ by an $O(1)$ shift of the arguments of Π, which by Proposition 2.8 leaves $\delta_{\mathfrak{G}}$ unchanged; we use (3.2) throughout.

Proposition 3.1. *[VERIFIED: BFS closure, exact] The group $\mathfrak{G} = \langle \mathfrak{a}_1, \mathfrak{a}_2, \mathfrak{a}_3, \mathfrak{b}, \mathfrak{h}\rangle$ generated by the involutions*

$$\begin{array}{ll} \mathfrak{a}_j\ (j=1,2,3): & \textit{swap rows } j \textit{ and } 4, \\ \mathfrak{b}: & \textit{swap columns } 3 \textit{ and } 4, \\ \mathfrak{h}: & (c_{11}\, c_{33})(c_{13}\, c_{31})(c_{22}\, c_{44})(c_{24}\, c_{42}), \end{array}$$

has order $1920 = |W(D_5)|$.

The order is computed by breadth-first closure on the sixteen-entry configuration and agrees with Zudilin's count. The group belongs to the Rhin–Viola series $W(A_4)$ of order 120 for $\zeta(2)$, $W(D_5)$ of order 1920 for $\zeta(3)$, $W(E_6)$ of order 51840 for $\zeta(4)$, and $\mathfrak{S}_7$ of order 5040 for the Brown–Zudilin $\zeta(5)$ cell integral [7].

For the hypergeometric parametrisation we pass to the variables of [17, §4]. With the normalisation $b_1 = 1$, set

$$h_0 = b_3 + b_4 - b_1 - a_1, \qquad h_{1,2,3} = 1 - b_1 + a_{2,3,4}, \qquad h_4 = b_4 - a_1, \qquad h_5 = b_3 - a_1. \quad (3.3)$$

In these variables the generators act by

$$\begin{aligned} \mathfrak{a}_j &: \ a_j \leftrightarrow a_4 \quad \text{(this moves } h_0\text{)}, \\ \mathfrak{b} &: \ b_3 \leftrightarrow b_4, \\ \mathfrak{h} &: \ (a_1, a_2, a_3, a_4; 1, b_2, b_3, b_4) \\ &\qquad \mapsto (b_3 - a_3,\ a_2,\ b_3 - a_1,\ a_4;\ 1,\ b_2 + b_3 - a_1 - a_3,\ b_3,\ b_3 + b_4 - a_1 - a_3). \end{aligned} \quad (3.4)$$

Remark 3.2 (why the first search fails)**.** The symmetric group $\mathfrak{S}_5$ permuting $h_1, \dots, h_5$ acts *trivially* on the rational function, because the function is a product over $j = 1, \dots, 5$. Hence the group is invisible at fixed h_0. The nontrivial elements are the row swaps $\mathfrak{a}_j$, and they *move* h_0. Any search for transfer identities inside this family must be cross-h_0; a search at fixed h_0 returns nothing and returns it for a trivial reason. This observation is worth recording because it is the reason a natural first search fails, and it also explains part of Finding 3.12.

3.2. **The transfer identity, verified coefficientwise.** Combining [17, eq. (4.4) and Lemma 9] gives the invariant in h-language. We have not seen it stated in this form, and derive it here:

$$\widetilde{I} := \frac{H(c)}{\Pi(c)} = \frac{\widetilde{G}(a,b)}{\prod_j (a_j - b_1)! \cdot \prod_j (a_j - b_2)!} = \frac{\widetilde{F}(h)}{\prod_{j=1}^5 (h_j - 1)! \cdot \big(1 + 2h_0 - \sum_j h_j\big)!}, \quad (3.5)$$

where

$$\widetilde{F}(h) = \sum_{t \ge 0} \widetilde{R}(h;t), \qquad \widetilde{R}(h;t) = (h_0 + 2t)\,(t+1)_{h_0 - 1} \prod_{j=1}^{5} \frac{1}{(t + h_j)_{1 + h_0 - 2h_j}}. \quad (3.6)$$

By Theorem 1, $\widetilde{F}(h) = A\zeta(3) + B$ with $A, B \in \mathbb{Q}$ computed exactly from the partial fractions of $\widetilde{R}$.

Proposition 3.3 (anchor)**.** *[VERIFIED: $n \le 4$, exact] At the Ball/Apéry point $h_0 = 3n+2$, $h_j = n+1$ one has $A/(2a_n) = 1, \frac{1}{4}, \frac{1}{36}, \frac{1}{576} = 1/n!^2$ for $n = 1,2,3,4$, where $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ is Apéry's numerator sequence* [1].

So the machinery reproduces Apéry's sequence exactly, confirming the $2\,n!^2 \sum \cdots$ normalisation of [2].

Finding 3.4 (the transfer identity)**.** [VERIFIED: 4 directions; orbits of sizes $1, 40, 80, 1920$; 1321 tested points; 0 failures] For every $g \in \mathfrak{G}$ the invariant (3.5) satisfies $\widetilde{I}(h) = \widetilde{I}(gh)$ *coefficientwise,* i.e. separately for A/N and for B/N, where N is the factorial normaliser of (3.5). The verification:

direction	$\lvert\mathfrak{G}(a,b)\rvert$	h-valid points	identity holds	fails	distinct κ
$(1,1,1,1;0,0,2,2)$ (symmetric)	1	1	1	0	1
$(2,2,3,3;0,1,4,5)$	40	40	40	0	3
$(3,4,4,5;0,2,7,7)$	80	80	80	0	5
$(4,7,8,11;0,3,13,14)$	1920	1200	1200	0	10

The 720 skipped points at the last direction are those at which a brick degenerates, $1+h_0-2h_j<1$; this is a direction-dependent boundary effect of the h-parametrisation, not a failure of the identity.

The ratio $\kappa = N(gh)/N(h) = \Pi(gc)/\Pi(c)$ is by construction a ratio of factorials. The identity itself is Bailey's/Whipple's transformation for very-well-poised ${}_7F_6$, [PROVED] in [17, Lemmas 7–9]; what Finding 3.4 adds is the exact coefficientwise check in the present normalisation.

Finding 3.5 (half-integer robustness — the p-adic point)**.** [VERIFIED: 3 directions; 780 comparable orbit points; 0 failures] Repeating the test with $h_4, h_5 \in \mathbb{Z}+\frac{1}{2}$ — testing proportionality of the full spectral vector by exact 2×2 determinants over every pole-residue class — gives

direction	comparable orbit points	proportional / not
$(2,2,3,3;0,1,4,5)$	120	120 / 0
$(3,4,4,5;0,2,7,7)$	180	180 / 0
$(4,7,8,11;0,3,13,14)$	480	480 / 0

The mechanism is lattice-robust: it is not an accident of integrality.

Finding 3.5 is what makes the Transfer Principle usable, because p-adic admissibility forces half-integer parameters at $p=2$ (Proposition 3.10). Without it, Corollary 2.5 would be a statement about a lattice the p-adic construction never visits.

3.3. The savings function.

Definition 3.6. With F the eight-element index set of $\Pi(c) = c_{21}!\,c_{31}!\,c_{41}!\,c_{12}!\,c_{32}!\,c_{42}!\,c_{33}!\,c_{44}!$, put

$$\nu_\ell = \max_{g\in\mathfrak{G}} \operatorname{ord}_\ell\big(\Pi(cn)/\Pi(gcn)\big), \qquad \Phi_n = \prod_{\sqrt{m_0 n}<\ell\le m_3 n} \ell^{\nu_\ell},$$
$$\Lambda(x) = \max_{g\in\mathfrak{G}} \sum_{i\in F}\big(\lfloor c_i x\rfloor - \lfloor (gc)_i x\rfloor\big), \qquad \delta_{\mathfrak{G}} = \int_0^1 \Lambda(x)\,\mathrm{d}\psi(x) - \int_0^{1/m_3} \Lambda(x)\,\frac{\mathrm{d}x}{x^2}, \tag{3.7}$$

ψ the digamma function, and

$$\text{budget} = 2m_1+m_2, \quad m_1 = \beta_4^*-\alpha_1^*, \quad m_2 = \max\{\alpha_1-\beta_1,\ \alpha_2-\beta_2,\ \beta_4^*-\alpha_3,\ \beta_4^*-\alpha_4,\ \beta_3^*-\alpha_1^*\}. \tag{3.8}$$

Finding 3.7 (end-to-end validation of the savings pipeline)**.** [VERIFIED: 8 digits, exact Λ, numerical quadrature of (3.7)] At Rhin and Viola's optimum $(\alpha;\beta) = (18,17,16,19;\,0,7,31,32)$ the implementation returns

$m_0 = 19$, $m_1 = 16$, $m_2 = 18$, $m_3 = 16$,
budget $= 50$, $\int_0^1 \Lambda\,\mathrm{d}\psi = 24.18768530$, $\int_0^{1/16} \Lambda\,\mathrm{d}x/x^2 = 4.00000000$,
$\delta_{\mathfrak{G}} = 20.18768530$, $C_2 = 29.81231470$, orbit of distinct F-multisets $= 118$,
$\mu(\zeta(3)) \le 5.51389063$ against the published 5.51389062.

Eight-digit agreement with a published irrationality measure that was obtained by entirely independent means is a strong joint test: the group, the invariant, the function Λ and both integrals in (3.7) are simultaneously confirmed.

3.4. **The totally symmetric point banks nothing.**

Proposition 3.8. *[PROVED] At the Apéry/Ball point $\alpha = (1,1,1,1)$, $\beta = (0,0,2,2)$ every entry of the matrix (3.2) equals $|\alpha_j - \beta_k| = 1$. A group acting by permutations of sixteen equal entries has a single orbit point, so the stabiliser is all of $\mathfrak{G}$ and*

$$\nu_\ell \equiv 0, \qquad \Phi_n \equiv 1, \qquad \delta_{\mathfrak{G}} = 0 \text{ exactly.}$$

The same holds at LSZ's $\zeta_2(5)$ point, where all bricks are again equal.

Proof. Every $g \in \mathfrak{G}$ is a permutation of the sixteen entries c_{jk}; if all entries are equal then $gc = c$ as a multiset and $\Pi(gcn) = \Pi(cn)$ for every g and every n. □

Proposition 3.8 is the good news and the bad news in one line. *The good news*: Apéry, Beukers, Calegari and LSZ have banked exactly zero group savings, so the orbit method is entirely fresh money. *The bad news* is §9.3: the locus where $\delta_{\mathfrak{G}} = 0$ is exactly the locus where the archimedean coefficient growth C_1 also vanishes, and the p-adic criterion must pay C_1.

3.5. **How much money is there.**

Finding 3.9. [VERIFIED: 9556 integral directions with $\sum \alpha \leq 34$; hill-climbing to $\sum \alpha \approx 250$] The scale-free relative saving $\delta_{\mathfrak{G}}/$ budget, maximised over integral directions satisfying (3.1):

directions	$\delta_{\mathfrak{G}}$	budget	$\delta_{\mathfrak{G}}/$ budget	$\delta_{\mathfrak{G}}$ at a budget $= 3$ configuration
$(4,7,8,11;0,3,13,14)$	11.9533	29	0.41218	1.2365
$(2,7,8,9;0,1,12,13)$	12.6730	32	0.39603	1.1881
Rhin–Viola optimum	20.1877	50	0.40375	1.2113
symmetric	0	3	0	0

The absolute ceiling found, subject to the construction still working, is $\delta_{\mathfrak{G}}/$ budget $= 0.4378$ at $(5,7,9,5;0,2,11,13)$.

Consistency with the literature: Brown and Zudilin's $\zeta(5)$ record point has $\delta_{\mathfrak{G}} = 34.394$ against $\sum m = 84$, i.e. 0.4094; Rhin and Viola's $\zeta(3)$ optimum gives 0.4038. Both land near 40%, and our independent maximisation caps out near 44%. It is fair to summarise: *a Rhin–Viola group removes about* 40% *of the denominator, fairly universally* [VERIFIED: the three directions above plus the hill-climb].

3.6. **Four honest flags.** The Transfer Principle transfers identities. It does not transfer admissibility, and it does not manufacture a group. Here are the four places where the p-adic setting differs, stated as they are, positive and negative.

Proposition 3.10 ((F1) very-well-poisedness forces $\theta = \frac{1}{2}$, i.e. $p = 2$)**.** *[PROVED] For $\Gamma(h_j + t)/\Gamma(1 + h_0 - h_j + t)$ to be a* rational *function of t one needs $2h_j - h_0 - 1 \in \mathbb{Z}$. With $h_0 \in \mathbb{Z}$ this forces $h_j \in \frac{1}{2}\mathbb{Z}$. Hence the only very-well-poised p-adic shift is $\theta = \frac{1}{2}$. For $p \geq 3$ one has $|\theta|_p \geq q_p$, so $\theta = a/p^l$, and the shifts must come as the* full *set $\{\nu/p^l : p \nmid \nu\}$ — which is exactly what* [13, Def. 16] *and* [11, §7] *do, and it is why those p-adic constructions abandon the factor $2t + h_0$.*

A sanity check on Proposition 3.10: Lai's normalising constant $p^{(l+1/(p-1))M\varphi(p^l)n}$ of [11, §7] collapses at $p = 2$, $l = 1$, $\varphi(2) = 1$, $M = 4$ to 2^{8n} — exactly LSZ's normalisation $C = 2^{8n}$. The case $p = 2$ is cheap precisely because $\varphi(2) = 1$.

Proposition 3.11 ((F2) admissibility constrains the base point only)**.** *[PROVED] For the Volkenborn integral at shift θ one needs*

(R1) *all poles of R at integers, else $R(t+\theta)$ has poles on $\mathbb{Z}_p$;*
(R2) *all numerator bricks at shift $\equiv -\theta \pmod 1$, else each numerator factor costs $-l$ instead of gaining $+1/(p-1)$ (see (4.6) below).*

Orbit points need satisfy neither. They enter only through (i) the exact rational transfer identity and (ii) their own denominator bound — both statements about $\mathbb{Q}$.

Proposition 3.11 is what makes the whole method conceivable, and it is the same logic as the Brown–Zudilin saving $\nu_p = \max_g \operatorname{ord}_p \prod_F h_i!/\prod_F (gh_i)!$ [7].

Finding 3.12 ((F3) the LSZ shape class has no internal transfer identity)**.** [EXCLUDED: exact rational arithmetic, zero tolerance; cross-h_0; ranges below] Parametrise the LSZ shape — numerator bricks at θ, denominator bricks at $\mathbb{Z}$, all centred, i.e. very-well-poised —

$$R(t) = (2t+h_0)\prod_{j=1}^{M}(t+\theta+e_j)_{h_0-2e_j}\Big/\prod_{j=1}^{M}(t+f_j)_{h_0+1-2f_j}.$$

An exhaustive exact search for proportional pairs of coefficient vectors — the exact signature of a transfer identity, by Corollary 2.5 — returns

family	points tested	proportional pairs
$M=2$ ($\zeta_p(3)$), all $h_0 \le 9$, cross-h_0	113	0
$M=4$ ($\zeta_p(5)$, LSZ), all $h_0 \le 7$, cross-h_0	1629	0
$M=4$, fixed $h_0=6$	672	0

One cannot find the group by deforming LSZ inside its own shape class: its group is trivial. This is a genuine negative, and it explains why the literature has not stumbled on the group.

Finding 3.13 ((F4) genuinely p-adic points do have transfer partners)**.** [VERIFIED: full very-well-poised half-integer family; 5507 distinct reduced rational functions, 598 genuinely p-adic; 13 proportionality classes] Widen the search to the full very-well-poised half-integer family ($h_0 \in \mathbb{Z}$, $h_j \in \frac{1}{2}\mathbb{Z}$, any q, reduced modulo brick cancellation). Of the 598 genuinely p-adic points — integer poles *and* a non-empty half-integer numerator — exactly 13 lie in a proportionality class containing a distinct partner. Four instances:

$$\begin{array}{llll}
h_0=5,\ \mathrm{num}\{\frac52,\frac52\},\ \mathrm{den}\{1^2,2^4,3^4,4^2\} & \leftrightarrow & h_0=4,\ \mathrm{den}\{1^4,2^4,3^4\} & \kappa=2\\
h_0=6,\ \mathrm{num}\{\frac52,\frac72\},\ \mathrm{den}\{2,3^4,4\} & \leftrightarrow & h_0=4,\ \mathrm{den}\{1^3,3^3\} & \kappa=1\\
h_0=7,\ \mathrm{num}\{\frac72,\frac72\},\ \mathrm{den}\{1,2,3^4,4^4,5,6\} & \leftrightarrow & h_0=5,\ \mathrm{den}\{1,2^4,3^4,4\} & \kappa=27/2\\
h_0=6,\ \mathrm{num}\{\frac52,\frac72\},\ \mathrm{den}\{1,2^3,3^4,4^3,5\} & \leftrightarrow & h_0=7,\ \mathrm{num}\{1,2,5,6\},\ \mathrm{den}\{3^4,4^4\} & \kappa=2^8
\end{array}$$

In every instance the partner is an *integer-parameter* point, i.e. a member of the classical archimedean family where the group and the whole denominator theory live.

The mechanism is a quadratic transformation — Gauss duplication, $(t+\frac12)_n(t+1)_n = 2^{-2n}(2t+1)_{2n}$ — the classical bridge between half-integer and integer very-well-poised parameters. This is the shape of the bridge one wants:

Open Lemma 3.14 (the Gauss-duplication bridge)**.** [OPEN] The $\theta = \frac12$ very-well-poised family maps, by Gauss duplication, onto the integer very-well-poised family, with an explicit factorial ratio κ; consequently a p-adic linear form equals κ times an integer-family linear form, and the integer family carries the full 1920-element orbit. *Status:* [VERIFIED: the 13 instances of Finding 3.13, at small size]; the systematic family version is not proved here, and it looks like a known quadratic transformation waiting to be quoted rather than a new theorem.

3.7. **Non-vanishing under orbit moves.** $\mathbb{Q}_p$ has no positivity, so non-vanishing of the linear form is a genuine constraint and must be tracked. Under an orbit move it is free.

Proposition 3.15. *[PROVED] An orbit move multiplies the whole linear form by the nonzero rational $\kappa = \Pi(gc)/\Pi(c)$. Hence the leading term is scaled and never cancelled; $S(gh) = 0 \iff S(h) = 0$, so an orbit move can neither create nor destroy vanishing; and the input to Bel's criterion — for LSZ, the Casoratian $\rho_{n,0}\rho_{n+1,3} - \rho_{n+1,0}\rho_{n,3} = 3 \cdot 2^{16n+18}/(n+1)^5$* [14, Lemma 15(b)] *— is scaled by $\kappa_n\kappa_{n+1} \neq 0$ and stays nonzero.*

Remark 3.16. The (F4) bridges of Finding 3.13 are *not* orbit moves: the partners are different rational functions. Non-vanishing must therefore be re-checked there. At every instance computed, $\kappa \neq 0$ (observed values $1, 2, 4, 16, \frac{27}{2}, 18, 54, \frac{4}{27}, \frac{4}{3}, 2^5, 2^8$), so no vanishing is introduced [VERIFIED: the 13 instances].

3.8. **Literature position: the empty intersection.**

Finding 3.17. [VERIFIED: cross-tabulation of a 37-paper corpus] A full cross-tabulation of the corpus for p-adic markers (`Volkenborn`, ζ_p, $\mathbb{Z}_p$) against group markers (`Rhin`, `permutation group`, `group of order`) returns *no paper that uses a Rhin–Viola group p-adically.* The intersection is empty. The two files that mention both state the matter as open: [14, Final remarks] and [6, Rem. 47], both quoted in §1.2.

The nearest existing object is [11, §§7–8], the only genuinely multi-parameter p-adic Volkenborn family, $V_n(t)$ with free $(M; \delta_1, \dots, \delta_J; l; w_n)$. Its saving is of the form $\Phi_n^{-s/J}$ with Φ_n an $\inf_y$ over floor functions — a Chudnovsky–Rukhadze–Hata brick saving, *not* a $\max_{g \in \mathfrak{G}}$ orbit saving. The two are independent and compose.

Finding 3.17 is the sense in which Theorem 1 is new: not that the underlying computation is deep, but that the intersection it inhabits was empty. The scope of the finding is exactly the corpus swept; we make no claim about the literature beyond it.

4. The parametrised construction and its exact cost ledger

The Transfer Principle says the group is available. Whether it is *worth* anything is a quantitative question, and to answer it one needs a cost model that is exact, that contains every published construction as a special case, and that is not fitted to them. This section builds it; §5 tests it.

4.1. **The family.**

Definition 4.1. Fix a prime p. The family is

$$R_n(t) = C^n \cdot (2t+n)^\epsilon \cdot \frac{\prod_{c=1}^{M} (t+\theta'_c)_{L_c}}{(t)_{n+1}^A}, \qquad S_n(\theta_0) = \int_{\mathbb{Z}_p} R_n^{(m)}(t+\theta_0)\,\mathrm{d}t, \qquad S_n = \sum_{\theta_0 \in \Theta} S_n(\theta_0), \tag{4.1}$$

with free parameter tuple

parameter	meaning	constraint
p	the prime	—
$\theta_0 = a/p^r$	integration shift, depth $r = -v_p(\theta_0)$	$\theta_0 \in \Theta$; see §4.3
θ'_c, $c \le M$	numerator Pochhammer shifts, depths l'_c	fractional; alignment §4.6
$L_c = \lambda_c n$	Pochhammer lengths (asymmetric allowed)	$\sum_c \lambda_c \le A$
A	pole order of $(t)_{n+1}^A$	$\deg R = \epsilon + \sum L_c - A(n+1) \le -2$
$m \ge 0$	derivative order	see §5.4
$\epsilon \in \{0,1\}$	the very-well-poised factor	$\epsilon \le A-2$ (degree)
C	normalisation — *forced*, §4.5	$C = p^{v_p(C)}$
Θ	cosets at which the *same* form must be small	§4.3

The asymmetric lengths $L_c = \lambda_c n$ are exactly the Rhin–Viola direction; setting all $\lambda_c = 1$ recovers the totally symmetric slice.

Finding 4.2 (the family contains the literature)**.** [VERIFIED: exact special cases with the printed constants] Definition 4.1 contains: [14, Def. 10] at $(p,r) = (2,1)$, $\theta' = \frac{1}{2}$ four times, $A = 4$, $m = 1$, $\epsilon = 1$, $C = 2^8$; Lai's family B_n [12, (3.2)] at $(2,2)$, $\theta' = \frac{3}{4}$ with multiplicity $s+2$, $A = s+2$, $m = s$, $\epsilon = 0$, $C = 2^{3s+6}$; Lai's family A_n [12, (3.1)] at $\theta' = \frac{1}{4}^{s+2}\frac{3}{4}^{s+2}$, $A = 2s+4$, $m = s$, $\epsilon = 1$, $C = 2^{6s+12}$; Beukers' $R^{(\mathrm{B})}$ [4] at $(2,1)$, $\theta' = \frac{1}{2}$ three times, $A = 3$, $m = 0$, $\epsilon = 1$, $C = 2^6$; and [13, Def. 16] together with [15, Def. 3.1] once free-factorial bricks are allowed (§4.10).

4.2. **What the form is a linear form in.**

Proposition 4.3. *[PROVED] Write $R_n(t) = \sum_{i=1}^{A}\sum_{k=0}^{n} r_{i,k}/(t+k)^i$ and set $J_u := \int_{\mathbb{Z}_p} \mathrm{d}t/(t+\theta_0)^u$ and $T_{k,u} := \sum_{\nu<k}(\nu+\theta_0)^{-u}$. Then*

$$S_n(\theta_0) = \rho_0 + \sum_{i=1}^{A}(-1)^m (i)_m\, \rho_i\, J_{i+m}, \qquad \rho_i = \sum_k r_{i,k}, \quad \rho_0 = -(-1)^m \sum_{i,k} r_{i,k}(i)_{m+1} T_{k,i+m+1}, \tag{4.2}$$

and $J_u = u\, p^{ru}\, \omega(a)^{-u}\, \zeta_p(u+1, \theta_0)$. Hence $S_n(\theta_0)$ is a $\mathbb{Z}_p$-linear form in 1 and the Hurwitz values $\zeta_p(i+m+1, \theta_0)$, $i = 1, \dots, A$.

This is [14, Lemma 5] and [13, Lemma 21] in the notation of Definition 4.1; it is the case $\theta = \theta_0$ of Theorem 1(ii) after m differentiations — except at $(p,r) = (2,1)$, where $|\theta_0|_2 = 2 < q_2$ and $\zeta_2(u+1, \frac{1}{2})$ is undefined: there Remark 2.3 applies and $J_u = u\, 2^{u+1}\zeta_2(u+1)$ by [14, Lemma 9], so the form is directly a form in 1 and the full ζ_2 values (this is regime S2).

Three mechanisms kill unwanted weights, each with its own parameter dependence.

- **(K1)** *purity by degree* [PROVED]: $\rho_1 = \lim_{t\to\infty} tR_n(t) = 0$ when $\deg R_n \le -2$ (Lemma 2.2). This kills the weight-$(m+2)$ term. It is the *only* mechanism that kills an odd weight.
- **(K2)** *even vanishing* [PROVED]: $\zeta_p(2k) = 0$. Available only for the full ζ_p, i.e. after the twisted coset sum, or at $(p,r) = (2,1)$ where [14, Lemma 9] makes the single shift $\frac{1}{2}$ already *be* the full ζ_2. Hurwitz values at even weight are not zero — as Lai notes, the values of $\zeta_2(\cdot, 1/4)$ at positive even integers are nonzero [12].
- **(K3)** *reflection* [PROVED], and [VERIFIED: 43 3-adic digits at $p = 3$]: $\zeta_p(s,x) = \zeta_p(s, 1-x)$, from $B_n(1-x) = (-1)^n B_n(x)$ and $\omega(-1) = -1$; so cosets pair up and only $\varphi(p^r)/2$ classes are independent.

Definition 4.4 (rank)**.** The *rank* of the form is the number of zeta values it carries:

$$\text{rank} = \#\{i \in [2,A] : i+m+1 \text{ odd}\} \text{ when (K2) applies}, \qquad \text{rank} = A-1 \text{ otherwise}. \tag{4.3}$$

Only rank $= 1$ yields an irrationality *measure* for an individual value; rank $= R > 1$ yields at best "one of R values is irrational".

Finding 4.5. [VERIFIED: against the literature] LSZ's $A = 4$, $m = 1$ gives rank $= 1$ at weight 5; $A = 6$, $m = 1$ gives rank $= 2$ at weights $\{5, 7\}$, which is Lai's "one of $\zeta_2(5), \zeta_2(7)$"; Lai's B_n gives rank $= s + 1$ over weights $[s + 3, 2s + 3]$, which is his Theorem 1.3.

4.3. Which cosets must be covered.

Proposition 4.6. *[PROVED] By (1.3), a form in* 1 *and the* full *value $\zeta_p(w)$ — not merely a Hurwitz value — requires either*

(a) $\varphi(p^r) = 2$*, so that a single Hurwitz value* is $\zeta_p(w)$ *up to a rational factor; or*
(b) *the full twisted sum $\Theta = \{j/p^r : p \nmid j\}$, which also switches on (K2), but then the smallness is a* minimum *over the $\varphi(p^r)$ cosets.*

We refer to case (a) as *regime S*, to the exceptional point $(p, r) = (2, 1)$ of (K2) as *regime S2*, and to case (b) as *regime T*. Which regime one is in is not a modelling choice; it is determined by (p, r). Case (a) is classified completely in §6.

4.4. Non-vanishing across the family.

$\mathbb{Q}_p$ has no positivity, so non-vanishing is a real constraint and it is *not* uniform over the family. We record the four mechanisms actually in use.

- **LSZ**: the exact Casoratian $\rho_{n,0}\rho_{n+1,3} - \rho_{n+1,0}\rho_{n,3} = 3 \cdot 2^{16n+18}/(n + 1)^5$ [14, Lemma 15(b)], [VERIFIED: exact, $n \le 11$]. It is available only because rank 1 and the three-term recursion both exist, i.e. only on the very-well-poised slice.
- **Lai** (A_n, B_n): no Casoratian; non-vanishing is proved only along $n = 2^m - 1$ by a dominant-term argument [12, Lemmas 2.4, 2.5, 6.1]. The cost is a subsequence.
- **Lai–Sprang / Lai–Lupu–Sprang**: the $\ell(n)$-adic criterion [13, Lemma 4] replaces non-vanishing by $v_{\ell(n)}(l_{0,n}) < v_{\ell(n)}(l_{i,n})$ with $\ell(n) = n+1$ prime. This *constrains the parameters*: it requires the extra numerator factor $(t + \theta_{\max})_{\theta_{\max}n-1}$ (resp. the factor t^{M_0} of [15]) and n in an arithmetic progression.
- **Group moves**: Proposition 3.15; free. The (F4) bridges are not, and must be re-checked (Remark 3.16).

4.5. The forced normalisation G.

Proposition 4.7. *[DERIVED] The coefficient $r_{i,k}$ carries $\prod_c \prod_j(\theta'_c - k + j)$, of p-denominator $p^{l'_c L_c}$, over $(k!(n - k)!)^A$, of p-denominator $p^{An/(p-1)}$. Hence p-integrality of the coefficients forces*

$$v_p(C) = \sum_c l'_c\lambda_c + \frac{A}{p - 1}, \qquad G := v_p(C) \log p. \tag{4.4}$$

Finding 4.8. [VERIFIED: every published prefactor] (4.4) reproduces exactly: 2^{8n} for LSZ $(4\cdot1+4)$; 2^{6n} for Beukers' $R^{(\mathrm{B})}$ $(3+3)$ and for Lai's B_n at $s = 0$ $(2\cdot2+2)$; $2^{(3s+6)n}$ and $2^{(6s+12)n}$ for Lai's two families; and $p^{pn}n!^s$ for [15], from $(p - 1) + (p - 1 + s)/(p - 1) = p + s/(p - 1)$.

Proposition 4.9 (G is the archimedean coefficient growth, on the co-located locus)**.** *[PROVED] in the sources. When the bricks are* balanced and co-located *— numerator and pole Pochhammers on the same interval — the hypergeometric factor is $e^{o(n)}$ and the archimedean growth of* $\max(|\rho_0|, |\rho_w|)$ *equals G. See* [4, Prop. 11.1(3)] *($|q_n|, |p_n| < e^{\varepsilon n}$, radius of convergence* 1*) and* [14, Lemma 20] *(characteristic polynomial $\lambda^2 - 2^9\lambda + 2^{16}$, with a* double *root 2^8).*

Finding 4.10. [VERIFIED: exact ρ's, growth measured] $\log\max|\rho|/n$ measured against G: $5.4958 \to G = 5.5452$ (LSZ); $4.15 \to 4.1589$ (Lai B_0); $8.21 \to 8.3178$ (Lai A_0); $3.276 \to 3.2958$ ($p = 3$).

Off the co-located locus the saddle contributes a positive $C_{\mathrm{sad}} = C_1 > 0$; this is measured in §4.9 and it is decisive in §9.3.

4.6. **The smallness exponent α.**

Proposition 4.11. *[DERIVED] from the $\triangle$-operator estimates $v_p(\int f) \geq \triangle(f) - 1$ and $\triangle(\binom{t+j}{n}) \geq -\log_p n$ of* [14, Lemmas 6–7]*, applied brick by brick to $R_n^{(m)}(t + \theta_0)$:*

$$\alpha = \Big[v_p(C) + A\,r + \sum_c \mathrm{ctb}_c(\theta_0)\Big]\log p, \quad \textit{minimised over } \theta_0 \in \Theta, \tag{4.5}$$

where

$$\mathrm{ctb}_c(\theta_0) = \begin{cases} +\dfrac{1}{p-1}, & \theta'_c + \theta_0 \in \mathbb{Z}_p \quad (\text{aligned}), \\ -\,l'', & \textit{otherwise, } l'' = -v_p(\theta'_c + \theta_0). \end{cases} \tag{4.6}$$

Equivalently, with unequal lengths (§4.9), $\alpha - G = (r + \frac{1}{p-1})\log p \cdot \big[\sum_d \nu_d - \sum_{\mathrm{mis}} \lambda_c\big]$.

Corollary 4.12 (the doubling)**.** *[DERIVED] When every copy is aligned and $l'_c = r$, (4.5) gives $\alpha = 2G$.*

Corollary 4.12 is LSZ's "totally symmetric hypergeometric" doubling, here derived rather than observed. It is also the exact point at which the archimedean and p-adic worlds part company: archimedeanly one has a saddle-point value; p-adically one has a valuation sum, and the sum happens to be twice the normalisation.

4.7. **The denominator cost E.**

Proposition 4.13. *[DERIVED] The denominator cost — the power of $d_n \sim e^n$ that must be cleared — is*

$$E = A + m + 1 - \epsilon. \tag{4.7}$$

The term $A + m + 1$ is the top pole order of $\sum_{i,k} r_{i,k}(i)_{m+1}(\nu + \theta_0)^{-(i+m+1)}$ combined with $d_n^{A-i} r_{i,k} \in \mathbb{Z}$ ([17, Lemma 16], [13, Lemmas 26–29]*); the $-\epsilon$ is the well-poised saving. When* rank $= 1$ *one has $E = w$, the top weight.*

Equation (4.7) is validated against every published anchor in §5.5 — and *corrected* there, in a range no published anchor visits.

4.8. **The criterion and the margin.**

Proposition 4.14 (Bel's criterion, as we use it)**.** *[PROVED]* [3, Lemme 3.2]*, quoted as* [14, Lemma 4]*. Set $a_n = \Phi_n^{-1} d_n^E \rho_0$ and $b_n = \Phi_n^{-1} d_n^E \rho_w$, integers, with $|a_n + b_n\xi|_p \leq e^{-\alpha n + o(n)}$ and $\max(|a_n|, |b_n|) \leq e^{\beta n + o(n)}$, $\beta = G + E$. Then*

$$\begin{gathered} \mathrm{margin} := \alpha - \beta \ \ (= G - E \ \textit{in the aligned case}), \\ \mathrm{margin} > 0 \implies \xi \notin \mathbb{Q} \quad \textit{and} \quad \mu(\xi) \leq \frac{\alpha}{\alpha - \beta} = \frac{2G}{G - E}. \end{gathered} \tag{4.8}$$

Remark 4.15 (the asymmetry that decides everything)**.** Compare with the archimedean Rhin–Viola criterion. There the irrationality condition is $C_0 > C_2$: the archimedean coefficient growth C_1 enters the *measure* $\mu \le (C_1 + C_2)/(C_0 - C_2)$ but never the irrationality condition. p-adically the coefficient *size* enters (4.8) directly through $\beta = G + E$. So the p-adic side pays for coefficient growth and the archimedean side does not. This one line is the whole of §9.3.

At the totally symmetric point, where $\alpha = 2G$ by Corollary 4.12, the ledger collapses to the three-line summary

$$\alpha = 2G, \qquad \beta = G + E, \qquad \text{margin} = G - E, \qquad \mu \le \frac{2G}{G - E}. \tag{4.9}$$

4.9. **Unequal lengths: where the saddle switches on.** Extending the ledger to unequal brick lengths in the manner of [17, (2.4)] — numerator bricks $(t + b_j)_{\lambda_j n}/(\lambda_j n)!$ and pole bricks $(\nu_j n - 1)!/(t + a_j)_{\nu_j n}$, with $\lambda_j = \alpha_j - \beta_j$ and $\nu_j = \beta_j - \alpha_j$ — gives

$$\alpha_p - \text{growth} = \Big(r + \frac{1}{p-1}\Big) \log p \cdot \Big[\sum_d \nu_d - \sum_{\text{mis}} \lambda_c\Big] - C_{\text{sad}}, \qquad C_{\text{sad}} = C_1. \tag{4.10}$$

Finding 4.16 (the saddle term, measured)**.** [VERIFIED: $p = 5$, $\theta_0 = 1/5$, exact, $n = 6, \ldots, 15$] Numerator bricks $(t + 4/5)_n$ and $(t + 4/5)_{2n}$ (lengths $\lambda = 1, 2$) over poles $(t)^3_{n+1}$, against the co-located control (lengths $1, 1, 1$):

	predicted $\alpha/\log 5$	measured $v_5(S_n)/n$	predicted G	measured growth	C_{sad}
unequal $\lambda = (1, 2)$	6.50	6.33–6.67	4.4260	5.77–5.86	$+1.35$
control $\lambda = (1, 1, 1)$	7.50	7.33–7.67	6.0354	5.97–6.06	≈ 0

The gauge-invariant margin rate closes exactly: $1.25 \log 5 \cdot 3 - 1.35 = 4.69$ against the measured $\alpha - \text{growth} = 6.50 \log 5 - 5.77 = 4.69$.

The saddle term switches on precisely when the bricks stop being co-located. That is the whole content of the tension in §9.3, and it is measured, not assumed.

4.10. **The free-factorial direction.** A numerator brick $n!^S$ in the Ball–Rivoal manner [2] has $l' = 0$, p-adic contribution $+1/(p-1)$ (since $v_p(n!) = n/(p-1)$) and archimedean cost $+\log 2$ per copy (the binomial saddle $\max_k \binom{n}{k} = 2^n$), and raises A, hence E, by one. So

$$\frac{\text{d margin}}{\text{d(free factorial)}} = \Big[\frac{1}{p-1} + r\Big] \log p - \log 2 - 1. \tag{4.11}$$

Finding 4.17 (the ledger reproduces the structure of the $p \ge 5$ theorem)**.** [DERIVED] + [VERIFIED: structure] Feeding [15, Def. 3.1] — $A = p - 1 + s$, $p - 1$ Pochhammer bricks at the shifts j/p, s free factorials, primitive so $E = A$ — through the ledger gives

$$\text{margin}(s) = (s + 1)\, \frac{p \log p}{p - 1} - s(1 + \log 2) - p + 1,$$

so a positive margin needs

$$s > \frac{p - 1 - \dfrac{p \log p}{p - 1}}{\dfrac{p \log p}{p - 1} - 1 - \log 2},$$

and the bracketed denominator is *exactly* the denominator of the constant c_p in [15, Thm. 1.1], namely

$$c_p = p + \frac{p-1-\varpi_p}{\dfrac{p\log p}{p-1} - 1 - \log 2}, \qquad \varpi_p = \psi(1/p) + 2p - 1 + \gamma + p\Big(\log p - \sum_{j\le p} \tfrac{1}{j}\Big).$$

The two numerators differ only in that the ledger's $p\log p/(p-1)$ is replaced by the exact quantity ϖ_p; the denominators, including the crude $\log 2$, agree exactly.

That a first-principles ledger reproduces the denominator of a published constant, in a paper written independently of it, is a strong test of the p-adic side of the model.

4.11. **The freedom map.** Which parameter moves which component:

direction	G	α	E	rank	net effect on margin
$A+1$	$+(r+\frac{1}{p-1})\log p$	$+2(r+\frac{1}{p-1})\log p$	$+1$	$+1$ per 2 steps	$+(r+\frac{1}{p-1})\log p - 1$, *but rank grows*
$r+1$	$+A\log p$	$+2A\log p$	0	0	$+A\log p$, *but $\varphi(p^r)$ cosets $\Rightarrow$ regime T*
$m+1$	0	0	$+1$	parity flip	-1; raises the weight by 1
$\epsilon: 0\to 1$	0	0	-1	0	$+1$ (needs $A\ge 3$)
misalign one copy	0	$-(r+\frac{1}{p-1})\log p$	0	0	$-(r+\frac{1}{p-1})\log p$
free factorial $+1$	$+\log 2 + \frac{\log p}{p-1}$	$+(\frac{2}{p-1}+r)\log p$	$+1$	$+\frac{1}{2}$	(4.11)
unequal λ_c	saddle on	linear in λ	via m_j	—	§4.9

It is worth being explicit about where the totally symmetric slice is *provably* optimal and where it is merely convenient, since this is the question the group was supposed to attack.

- **Provably optimal.** (i) *Alignment*: every copy aligned maximises $\sum_c \mathrm{ctb}_c$, since $\mathrm{ctb}_c \le 1/(p-1)$ termwise [PROVED]; so on the single-coset regimes the symmetric point maximises $\alpha - G$ at fixed A, r. (ii) *Balance and co-location*: this is exactly the locus where the hypergeometric saddle vanishes, $C_{\mathrm{sad}} = 0$ [PROVED] by [4, Prop. 11.1(3)] and [14, Lemma 20], which is what makes the growth equal G.
- **Merely convenient.** (iii) The *equal exponents* $L_c = n$ are not forced; unequal lengths are exactly the Rhin–Viola freedom and are what a positive $\delta_{\mathfrak{G}}$ requires. (iv) $\epsilon = 1$ and its E-saving are conveniences of the well-poised shape. (v) LSZ's $(m, A) = (1, 4)$ is one of two rank-1 routes to weight 5 at $p = 2$; $(m, A) = (2, 3)$ also works, with the worse margin $6\log 2 - 5 < 0$.
- **Provably not optimal.** Nothing in the symmetric slice fixes r: at $p = 2$ the depth-2 point (Beukers $F = 4$ / Lai B_n) and the depth-1 point (Beukers $R^{(\mathrm{B})}$) give the *identical* measure — a genuine one-parameter degeneracy of the ledger, and, as [14, Final remarks] observe from the other side, the two forms coincide.

4.12. **The exported margin function.** For reference, the entire model is the following six lines.

$$\begin{aligned}
\mathrm{margin}(p, r, \mathrm{shifts}, A, m, \epsilon) &= \Big[A\,r + \min_{\theta_0\in\Theta} \sum_c \mathrm{ctb}_c(\theta_0)\Big]\log p - E,\\
E &= A + m + 1 - \epsilon,\\
\Theta &= \{1/p^r\} \quad \text{if } \varphi(p^r) = 2 \text{ or } (p, r) = (2, 1) \qquad \text{(regimes S, S2)},\\
\Theta &= \{j/p^r : p \nmid j\} \quad \text{otherwise} \qquad \text{(regime T)},\\
\mathrm{rank} &= \#\{i \in [2, A] : i + m + 1 \text{ odd}\}\ \text{(S2, T)}, \qquad \mathrm{rank} = A - 1\ \text{(S)},\\
w &= \text{the surviving odd index;} \quad \mathrm{rank}\ \text{must be 1 for a measure.}
\end{aligned} \tag{4.12}$$

Its inputs are the parameter tuple and nothing else. Everything in §5, §7 and §8 is computed from (4.12) plus the corrections of §5.5.

5. Validation

Everything after this section is a computation inside the model of §4. The model's credibility therefore rests here, and we make the test as unforgiving as we can: the inputs are only the parameter tuples read off the sources, and the outputs are compared with printed constants that were obtained by entirely different routes.

5.1. **The gate.**

Finding 5.1 (all three published measures, from the tuple alone)**.** [VERIFIED: exact; input is only $(p, r, \text{shifts}, A, m, \epsilon)$] The ledger (4.12) derives E, the regime, the weights, the rank, G, α, β and μ, and returns:

target	parameters	regime	rank	E	G	$\mu \le$
$\zeta_2(3)$ (Beukers $F = 4$ / Lai B_n, $s = 0$)	p=2, r=2, A=2, m=0, ϵ=0	S	1	3	$6 \log 2$	7.17739889912418
$\zeta_2(3)$ (Beukers $R^{(\mathrm{B})}$)	p=2, r=1, A=3, m=0, ϵ=1	S2	1	3	$6 \log 2$	7.17739889912418
$\zeta_3(3)$ (Beukers $F = 3$)	p=3, r=1, A=2, m=0, ϵ=0	S	1	3	$3 \log 3$	22.28144795149432
$\zeta_2(5)$ (LSZ)	p=2, r=1, A=4, m=1, ϵ=1	S2	1	5	$8 \log 2$	20.34265173891448

against the published $7.177398\ldots$, $22.281447\ldots$, $20.342651\ldots$ of (1.1). Every printed digit matches.

5.2. **Four further independent anchors.** A model that reproduces three numbers with three parameters is not yet evidence. The following four checks use no additional freedom.

(i) **Beukers' second $\zeta_2(3)$ form.** Rows 1 and 2 of Finding 5.1 are *different* parameter points — depth 2 with $\epsilon = 0$, and depth 1 with $\epsilon = 1$ — and the ledger returns the identical measure. This matches LSZ's own remark that the two forms coincide [14, Final remarks]. [VERIFIED: exact]

(ii) **Lai's family B_n, for every s.** The ledger returns both the smallness rate $(6s + 12) \log 2$ and the margin $(3 \log 2 - 2)s + 6 \log 2 - 3$, matching the printed constants of [12]. [VERIFIED: $s = 0, 1$ printed; formula for all s]

(iii) **Lai's family A_n, for every s.** The ledger returns the smallness rate $(10s + 20) \log 2$, matching [12]. Here the ledger is *sharper* than the source on the archimedean side — Lai prints $(2 + 4 \log 2)s + 4 + 8 \log 2$, the ledger gives $(6s + 12) \log 2$ — and the exact measurement $8.12 \to 8.21$ at $n = 8 \to 24$ confirms the ledger, converging to 8.3178. [VERIFIED: exact, $n \le 24$]

(iv) **The denominator of the $p \ge 5$ constant.** Finding 4.17: the ledger reproduces the exact denominator of c_p in [15, Thm. 1.1]. [DERIVED]

On the three quantities the ledger models — the smallness rate α, the forced normalisation G and the denominator exponent E — there is no case in which a source is sharper, and one case (iii) in which the ledger is sharper than a printed bound. On the *margin* there is one exception, and it is the one predicted by §5.6(a): for Lai's A_n at $s \ge 1$ his printed margin $(6 \log 2 - 4)s + 12 \log 2 - 7$ exceeds the ledger's $(4 \log 2 - 3)s + 8 \log 2 - 4$, because his family carries an additional asymptotic $\Phi^{-(s+2)}$ that the ledger has no term for. At $s = 0$ the ledger is the sharper of the two.

5.3. **Independent re-derivation of the LSZ anchor.**

Finding 5.2. [VERIFIED: exact rational arithmetic; ranges stated per item] Starting from the partial fractions of [14, Def. 10] alone, we re-derive: their printed $\rho_{0,3} = 768$, $\rho_{1,3} = 73728$, $\rho_{0,0} = 0$, $\rho_{1,0} = -1024$; the vanishing $\sum_k r_{n,1,k} = 0$; the three-term recursion for both rows ($n \le 11$); the Casoratian $3 \cdot 2^{16n+18}/(n+1)^5$ ($n \le 11$); and the inclusions $\rho_{n,3} \in \mathbb{Z}$, $d_n^5 \rho_{n,0} \in \mathbb{Z}$ ($n \le 12$). Independently, the normalisation constants $\rho_n = 1, 96, 14944$ are reproduced. Cross-checking the p-adic side against an independently validated implementation of the Kubota–Leopoldt p-adic L-function [10], the reflection $\zeta_3(3, 1/3) = \zeta_3(3, 2/3)$ holds to 43 3-adic digits.

Note the third item in the inclusions: the measured d_n-multiplicity clearing ρ_0 for LSZ is 5, which is their *conjectured* d_n^5 of the "(den-con)" discussion [14, Final remarks], not merely the d_n^6 they prove. The same happens for Lai's A_n, where the measured $E = A + m + 1 - \epsilon = 3s + 4$ is one better than his proved d_n^{3s+5}.

5.4. **The direction that looked free and is not.** The ledger's α in (4.5) does not depend on m, while $E = A+m+1-\epsilon$ is smallest at $m = -1$ (the primitive, i.e. the Lai–Sprang convention). Taken at face value, applying this to *the very rational function LSZ use*, $2^{8n}(2t+n)(t+\frac{1}{2})_n^4/(t)_{n+1}^4$, would give a rank-1 form in 1 and $\zeta_2(3)$ with $E = 3$, $G = 8\log 2$, margin $+2.5452$ and $\mu(\zeta_2(3)) \le 4.3574$ — a large record. It is false, and it is worth seeing why, because the failure is structural and it disciplines the rest of the scan.

Finding 5.3 ($m = -1$ is dead)**.** [VERIFIED: exact truncated Volkenborn, $n \le 22$] Measured smallness rates $v_2(C^n S_n)/n$ against the ledger prediction:

family	predicted α	measured $v_2(C^n S_n)/n$
$M = 3$, $m = 0$ (Beukers $R^{(\mathrm{B})}$)	$12 \log 2$	$11.50 \ldots 11.83\ \log 2$
$M = 4$, $m = 1$ (LSZ)	$16 \log 2$	$15.17 \ldots 15.72\ \log 2$
$M = 5$, $m = 0$	$20 \log 2$	$19.17 \ldots 19.50\ \log 2$
$M = 6$, $m = 1$	$24 \log 2$	$23.14 \ldots 23.50\ \log 2$
$M = 4$, $m = -1$	$16 \log 2$	$\mathbf{8.11 \ldots 8.33\ \log 2}$
$M = 6$, $m = -1$	$24 \log 2$	$\mathbf{11.90 \ldots 12.14\ \log 2}$

That is, $\alpha = G$, not $2G$: *the primitive loses the entire p-adic doubling*, so margin $= -E < 0$ always. Confirmed a second way with no Volkenborn truncation at all: $y_n := -\rho_0/\rho_w$ converges 2-adically — so the linear-form identity is right — but at exactly half the rate of the $m \ge 0$ controls (135 versus 272 digits at $n = 18$).

Proposition 5.4. *[PROVED] The smallness of $\int R^{(m)}(t+\theta_0)\mathrm{d}t$ comes from the* product *structure: $v_2(C^n R(t+\frac{1}{2})) = 2Mn + Mn + Mn = 4Mn$ is an exponential cancellation between partial-fraction pieces that are individually of size 2^{-2Mn}. The primitive $\widetilde{R}$ is not that product; its pieces $r_{i,k}(t+\frac{1}{2}+k)^{1-i}/(1-i)$ are added without cancellation, and $v_2(C^n\widetilde{R}) = O(1)$.*

Correction 1. The $m = -1$ slice is not a slice of the construction: $\alpha(m = -1) = G$. The ledger must carry $m \ge 0$, and (4.12) is stated with that constraint.

5.5. **The denominator cost, measured — and corrected.** Equation (4.7) is validated against LSZ ($m = 1$), Beukers ($m = 0$), Lai's B_n and A_n ($m = 0, 1$). *Every anchor has $m \le 1$.* But weight 7 at rank 1 forces $m \ge 3$ (§7.4), so the ledger extrapolates into unmeasured territory exactly where the next open question lives. We measured it.

The exact denominator of ρ_0 was factored — after applying the forced normalisation $C^n = p^{v_p(C)n}$, with the p-part removed since C^n pays it — and the exponent read off as a function of ℓ/n. The profile is a clean two-band step:

$M=4$ very-well-poised, $p=2$, $r=1$, $n=48$			
$m=1$	exponent 5 on every prime $\ell \le n$	nothing above n	$E=5$ (ledger 5)
$m=3$	exponent 7 on every prime $\ell \le n$	exponent 4 on $n<\ell<2n$	$E=11$ (ledger 7)
$m=5$	exponent 9 on every prime $\ell \le n$	exponent 6 on $n<\ell<2n$	$E=15$ (ledger 9)

Since $\log \operatorname{lcm}\{\ell : n<\ell<2n\}/n \to 1$, the second band costs its exponent outright.

Finding 5.5 (two denominator rules)**.** [VERIFIED: $p \in \{2,3,5,7\}$, $r \in \{1,2\}$, $A \le 5$, $m \le 4$, $\epsilon \in \{0,1\}$, $n = 32,\dots,48$; 0 violations in 1084 non-degenerate configurations]

(R1): The well-poised saving $-\epsilon$ is real *only where the symmetry* $R(-h_0-t) = \pm R(t)$ *actually holds*: at $(p,r)=(2,1)$ with every shift $\frac{1}{2}$, and at $(2,2)$ with the $\frac{1}{4}$- and $\frac{3}{4}$-bricks *paired* (Lai's A_n). Everywhere else — $p \ge 3$, and $(2,2)$ with all bricks on one coset — the measured exponent on the primes $\le n$ is $A+m+1$ exactly: the factor $2t+n$ buys nothing.

(R2): A second band appears on the primes in $(n,2n)$, of exponent $c=m+1$ when $m \ge A-1$, and $c=0$ otherwise (single-shift shape).

Hence

$$E_{\text{true}} = (A+m+1-\epsilon_{\text{eff}}) + c. \tag{5.1}$$

The band comes from the half-integer sums $T_{k,u} = \sum_{\nu<k}(\nu+\theta_0)^{-u}$, whose denominators run over the odd numbers up to $2n$: for small m they cancel against the $r_{i,k}$, and for $m \ge A-1$ they do not.

Remark 5.6 (the one exception, at the bottom of the range)**.** At $n=30$ the single configuration $(p,r,A,m,\epsilon) = (5,2,5,2,1)$ reads exponent 7 on the primes $\ell \le n$ where (R1) predicts 8; the plateau has not yet formed there (the exponents are $7,7,7,8,8,7,8$ on $\ell = 7,11,13,17,19,23,29$), and the same configuration reads 8 at $n = 32,34,36,42,48$. This is why the range above starts at $n=32$.

No published result is touched. All four anchors of Finding 5.1 have $c=0$ and the correct ϵ_{eff}, and the measurement re-derives their $E=3,3,5,3$ exactly. Lai's B_n at $s=2,3$ (i.e. $m=2,3$) also has $c=0$, since $m<A-1$ there, so his printed denominators stand as well.

Finding 5.7 (the band cannot be shaped away)**.** [COMP: the whole inset lattice $[0,2]^4 \times [0,2]^4$ at the rank-1 $\zeta_2(7)$ point $M=4$, $m=3$, $\epsilon=1$] The attainable pairs (exponent on $\ell \le n$, exponent on $n<\ell<2n$) are $(6,6),(7,4),(7,5),(7,6),(7,7),(8,8)$. The minimum second-band exponent is 4, attained at the symmetric point, so $E_{\text{true}}=11$ is optimal over the lattice.

Correction 2. E_{true} at the rank-1 $\zeta_2(7)$ point is 11, not 7; the margin is -5.4548, not -1.4548. Likewise $\zeta_2(9)$: -9.4548; $\zeta_3(5)$: -3.8028; $\zeta_3(7)$: -8.7042.

Correction 3. The $\epsilon=1$ saving is a $p=2$ phenomenon. Granting it at every p, as a naive reading of (4.7) does, makes every $p \ge 3$ cell that uses it 1 better than the truth.

All three corrections make the picture *worse*, and none touches a published anchor. That combination is the reason we regard the ledger as validated rather than tuned: a fitted model does not produce corrections that hurt only in the unmeasured range.

The status of (R2) deserves a sentence of its own, since it is one of the two open lemmas of §9.1. It is [VERIFIED: 0 violations over the range stated in Finding 5.5], it is not [PROVED], and it was verified for the single-shift shape; the multi-coset spreads at $p \ge 5$ inherit it *by assumption.* As we note in §8.2, the $p \ge 5$ conclusions do not depend on it, those cells being 3 to 15 short either way.

5.6. **What the ledger does not capture.** For completeness, three known gaps.

(a) *Family-specific* Φ_n *savings.* $E = A+m+1-\epsilon$ is the *measured* multiplicity for the anchors, but e.g. Lai's A_n additionally carries an asymptotic $\Phi^{-(s+2)}$. A sharper E would move every margin by $O(1)$; it would not change any sign in §8, where the deficits are 3 to 15.

(b) *Coset cancellation.* $\alpha = \min_j$ is an inequality and the ledger has no term for cancellation between cosets. This is the escape hatch, tested in §7.8.

(c) *The saddle off the co-located locus* is modelled by $C_{\text{sad}} = C_1$ and measured in Finding 4.16, but only at the directions tested.

6. The classification of two-term forms

The ledger has one input that is not a real number and cannot be optimised away: the number of cosets. This section shows that it decides the shape of the whole problem.

6.1. $\varphi(p^r) = 2$.

Theorem 2. *[PROVED] Within the family of Definition 4.1, a two-term* $\mathbb{Z}_p$*-linear form in* 1 *and the* full *value* $\zeta_p(w)$ *— as opposed to a form in* 1 *and a Hurwitz value* $\zeta_p(w, \theta_0)$ *— arises from a* single *integration shift if and only if*

$$\varphi(p^r) = 2, \qquad \textit{i.e.} \qquad p^r \in \{3, 4\}, \qquad \textit{i.e.} \qquad (p, r) \in \{(3, 1), (2, 2)\},$$

together with the exceptional point $(p, r) = (2, 1)$*, at which* [14, Lemma 9] *makes the single shift* $\frac{1}{2}$ *already* be *the full* ζ_2*. In every other case the full twisted coset sum* $\Theta = \{j/p^r : p \nmid j\}$ *is forced, and the p-adic smallness of the resulting form is* $\min_{\theta_0 \in \Theta}$ *of the per-coset smallness.*

Proof. By (1.3) ([13, Lemma 12]), for $q_p \mid D$ and $i \geq 2$ one has $D^i \zeta_p(i) = \sum_{p \nmid j} \omega(j/D)^{1-i} \zeta_p(i, j/D)$, a sum over $\varphi(D)$ terms. By the reflection (K3), $\zeta_p(s, x) = \zeta_p(s, 1-x)$, these pair up into $\varphi(D)/2$ independent classes. A single Hurwitz value is therefore a rational multiple of $\zeta_p(i)$ precisely when $\varphi(D)/2 = 1$, i.e. $\varphi(p^r) = 2$, i.e. $p^r \in \{3, 4\}$. Concretely these are Beukers' identities $T_2(1/4) = 4^3 \zeta_2(3)$ and $T_3(1/3) = 3^3 \zeta_3(3)$ [4, Prop. 5.2]. The exceptional point is [14, Lemma 9]: $\frac{1}{s-1} \int_{\mathbb{Z}_2} \mathrm{d}t/(t + \frac{1}{2})^{s-1} = 2^s \zeta_2(s)$, in which the single shift already produces ζ_2 itself. Otherwise a single Hurwitz value spans a proper subspace and one must sum over Θ; the p-adic valuation of a sum is at least the minimum of the valuations, with equality unless there is cancellation (tested in §7.8). □

Theorem 2 is a hard finiteness statement: exactly three (p, r)-points in the entire family produce a two-term form for a single value from a single shift, and they are exactly the three points at which the three known measures (1.1) live. It follows that any attack on $\zeta_p(w)$ with $p \geq 5$ is *necessarily* in regime T.

6.2. At $p \geq 5$ the smallness cancels the growth exactly.

Proposition 6.1. *[DERIVED], and [VERIFIED: numerically at* $p = 5$*, exact] For* $p \geq 5$*, every rank-*1 *configuration of Definition 4.1 is in regime T. If moreover* $\varphi(p^r)$ *exceeds the number of distinct numerator-brick shifts — so that some coset carries no aligned brick — then at that coset*

$$A\, r + \sum_c \operatorname{ctb}_c(\theta_0) = A\, r - A\, r = 0 \quad \textit{exactly,}$$

whence $\alpha = G$ *and* margin $= -E$*. That is:* the p-adic smallness of the twisted sum exactly cancels the archimedean growth of its coefficients, and the entire d_n^E denominator cost is unpaid.

Proof sketch. In regime T the shifts must be the full set $\{\nu/p^r : p \nmid \nu\}$ (Proposition 3.10), and α is a minimum over $\theta_0 \in \Theta$. A numerator brick at θ'_c is aligned with exactly one coset, namely $\theta_0 \equiv -\theta'_c$. Under the hypothesis some θ_0 has no aligned brick; at that coset each of the A bricks contributes $-l'' = -r$ by (4.6), cancelling the $+Ar$ of (4.5), so $\alpha = v_p(C)\log p = G$ and margin $= \alpha - G - E = -E$. □

Remark 6.2 (what happens without the hypothesis)**.** [VERIFIED: exact, $p = 5$] The hypothesis is not vacuous and it is not automatic. It holds at the two cells that carry the headline of §8.2 — $\zeta_5(3)$ and $\zeta_7(3)$, where $A = 2 < \varphi(p)$ — and there margin $= -E = -3$ exactly. It fails as soon as the bricks can be spread over all cosets: at $p = 5$, $r = 1$, $A = 4$, $m = 1$, $\epsilon = 1$ with the four bricks on the four cosets $j/5$, the worst coset still carries one aligned brick, the contributions are $(-1, -1, -1, +\frac{1}{4})$, and

$$\text{margin} = \left(4 - \tfrac{11}{4}\right)\log 5 - E = \tfrac{5}{4}\log 5 - 5 = -2.9882,$$

which is the $(p, w) = (5, 5)$ entry of Finding 8.1. In general the correct statement is the ceiling of Corollary 6.4, margin $\le A\, p\log p/(p-1)^2 - E$, which covers every A, r and m in regime T and is what the $p \ge 5$ verdict rests on.

Finding 6.3 (the alignment law made visible)**.** [VERIFIED: exact $v_p(S_n)$; configurations and n stated] Predicted $\alpha/\log p$ against measured $v_p(S_n)/n$:

configuration	predicted $\alpha/\log p$	measured $v_p(S_n)/n$
LSZ, $\zeta_2(5)$	16	16.00 ($n = 8, 16, 32$)
Lai B_n, $s = 0$ / $s = 1$	12 / 18	12.06 / 18.25 ($n = 16$); Lai prints 12, 18
Lai A_n, $s = 0$ / $s = 1$ (mixed alignment)	20 / 30	20.04 / 30.1 ($n = 24$); Lai prints 20, 30
Beukers $p = 3$, weight 3	6	$5.88 \to 6$ ($n = 32$)
$p = 5$, 4 bricks split over 2 cosets	7.5	7.50 ($n = 12$)
$p = 5$, 4 bricks all on coset 1/5, *evaluated at* 2/5	$5 = v_5(C)$	5.33 ($n = 12$) $\Rightarrow$ rate 0

The last line is Proposition 6.1 in one row: the same form is p^{-10n}-small at one coset and only p^{-5n}-small at the other, and $5 = v_5(C)$ means the p-adic smallness has been exactly cancelled by the archimedean growth.

Corollary 6.4 (raising A does not help)**.** *[DERIVED] The aligned fraction is at most $1/c_p$ with $c_p = \varphi(q_p)/2$, so*

$$\text{margin} \le A \cdot \frac{p\log p}{(p-1)^2} - E,$$

and $p\log p/(p-1)^2 < 1$ for every $p \ge 3$, with values 0.823959, 0.502949, 0.378371, 0.263768, 0.231558 *at $p = 3, 5, 7, 11, 13$, while $E \ge A$. Raising r multiplies the number of cosets and makes matters worse. Hence* positive margins at $p \ge 5$ exist only at rank ≥ 2*, which is exactly the "one of $\zeta_p(3), \ldots, \zeta_p(c_p)$" regime of* [15] *— never an individual value, never a measure.*

6.3. **What is actually missing at $p \ge 5$.** It is worth being precise about the nature of the $p \ge 5$ obstruction, because it is not what one might guess.

Remark 6.5. Beukers already proves the *individual Hurwitz values* irrational for every $p \ge 3$: $\omega(a)^{-2}H_p(3, a, p) \notin \mathbb{Q}$, [4, Cor. 11.3], his condition (C) holding for every prime power $F > 2$. What is missing at $p \ge 5$ is therefore not smallness of a p-adic linear form in *some* Hurwitz value; it is *simultaneity.* By [4, Prop. 5.2],

$$T_5(1/5) = \tfrac{5^3}{2}\big(\zeta_5(3) - L_5(3, \chi_5)\big), \qquad T_5(2/5) = \tfrac{5^3}{2}\big(\zeta_5(3) + L_5(3, \chi_5)\big),$$

so $\zeta_5(3)$ sits in a two-dimensional space spanned by two Beukers-irrational numbers and is their half-sum. Beukers' machine proves each of $T_5(1/5), T_5(2/5)$ irrational; it does not prove their half-sum irrational, and one rational function can be p-adically aligned with only one coset at a time.

This is the sharpest statement we can make of the $p \geq 5$ problem: it is a *simultaneous* approximation problem in disguise, and the family of Definition 4.1 is a rank-1 instrument.

7. The scan

With the ledger validated and the classification in hand, the remaining question is empirical: does anything in the family beat what is known, and if not, exactly what stops it? This section reports three computations — an optimality computation, a rank-reduction search, and a coset-cancellation sweep — each with its domain, its tolerance and, where applicable, its positive control.

7.1. **Four doors that had to be re-opened.** A first optimiser over (4.12) necessarily leaves gaps, and it is worth naming them, because two of the four changed the answer.

(1) $m = -1$, the primitive convention, is not a valid slice: Finding 5.3.

(2) E at $m \geq 2$ had never been measured, every anchor having $m \leq 1$: Finding 5.5. This changed the $\zeta_2(7)$ answer by 4 units.

(3) The *rank* axis. Reporting only rank-1 cells hides the fact that at $p = 2$ the size constraint is never binding: §7.4.

(4) The $O(1)$ *inset* cone — bounded translations of the bricks. These move no asymptotic rate at all, so they are invisible to the ledger, but they move every rational coefficient, and so they are exactly where a rank reduction could hide: §7.5.

Before anything else, the scan code reproduces the anchors of §5 independently: LSZ's printed $\rho_{0,3} = 768$, $\rho_{1,3} = 73728$, $\rho_{0,0} = 0$, $\rho_{1,0} = -1024$, the normalisations $\rho_n = 1, 96, 14944$, and all three published measures to every digit [VERIFIED: exact].

7.2. **The optimisation domain.** We state the domain first, so that the theorem that follows is unambiguous.

Definition 7.1 (the cone)**.** The *cone* $\mathcal{C}$ consists of the rational functions

$$R(t) = (2t + h_0)^{\epsilon} \, \frac{\prod_{j \leq M} (t + \theta_j' + e_j)_{h_0 - 2e_j}}{\prod_{j \leq M} (t + f_j)_{h_0 + 1 - 2f_j}} \tag{7.1}$$

obtained from Definition 4.1 by allowing, in addition to the parameter tuple $(p, r, \text{shifts}, A, m, \epsilon)$ with

$$p \leq 13, \quad r \leq 3, \quad A \leq 14, \quad 0 \leq m \leq w + 2, \quad \epsilon \in \{0, 1\}, \quad \text{all coset spreads}, \quad w \leq 9,$$

the following brick displacements:

(a) $O(1)$ *insets*: integer $e_j, f_j \geq 0$ bounded independently of n (58, 127 and 288 admissible points at $M = 3, 4, 5$ respectively);

(b) *scaled insets*: $e_j = \eta_j n$, $f_j = \phi_j n$ with $\eta_j, \phi_j \geq 0$ — this is the Rhin–Viola length cone, and it is where $\delta_{\mathfrak{G}} > 0$ lives;

(c) *the group orbit*: the whole 1920-element $\mathfrak{G}$-orbit of the direction, via Corollary 2.5;

(d) *rank reduction*: all $\mathbb{Q}$-linear combinations $L = \sum_{j,l,d} c_{j,l,d} \, n^d \, (n \to n + l)$ (inset-form j) with polynomial coefficients, within the shift/degree bounds of §7.5.

Negative pole insets are excluded throughout, since they put poles on the wrong side of $\mathbb{Z}_p$.

Remark 7.2 (no calibration step)**.** In the computations of this section E and the archimedean growth are *measured*, not taken from (4.7) and (4.4). Consequently any Φ_n-type or Rhin–Viola saving that is actually present at a point of $\mathcal{C}$ is already counted in the number reported for that point. There is no fitting step anywhere in §7.

7.3. **Optimality of the three known measures.**

Theorem 3. *[COMP: the cone $\mathcal{C}$ of Definition 7.1] Over $\mathcal{C}$, the three published p-adic irrationality measures are attained and cannot be improved:*

$$\mu(\zeta_2(3)) \le 7.17739889912418, \qquad \mu(\zeta_3(3)) \le 22.28144795149432,$$
$$\mu(\zeta_2(5)) \le 20.34265173891448,$$

and no configuration in $\mathcal{C}$ yields a smaller bound for any of the three, nor a finite bound for any other $\zeta_p(w)$ with $p \le 13$, $w \le 9$.

We emphasise, as in §1.4: this is optimality *over* $\mathcal{C}$. It is not a statement that these bounds are unimprovable.

The evaluation is of $\mu = \alpha/(\alpha - \beta) = (G + \mathrm{gain})/(\mathrm{gain} - C_1 - E)$ with E and the growth measured. The four sub-computations:

(i) $O(1)$ **insets: asymptotically no effect, and never an improvement.** At all 58/127/288 admissible points ($M = 3, 4, 5$) every *asymptotic* quantity — $\sum\lambda = \sum\nu = M$, G, the gain, $C_1 = 0$ — is identical to the symmetric point, and the measured E is identical too, so μ is asymptotically unchanged, exactly. At finite n the insets perturb $\sum\lambda, \sum\nu$ by $O(1/n)$, and the symmetric point $e = f = 0$ is the best point of the cone at every n tested, at both rank-1 points; at the rank-2 control $M = 5$ it is the best point at $n \ge 36$ but not at $n = 24$, for the artefactual reason recorded next. [VERIFIED: 473 points; exact; $n = 24$, spot-checked at $n = 36, 48, 60$] *A caveat honestly recorded*: a promising drop $E : 5 \to 3$ at $M = 5$, $n = 24$ was chased and is a small-n artefact; at $n = 36, 48, 60$ the exponent is back to 5. See Appendix A for a second finite-n artefact in the same printout.

(ii) **Scaled insets: strictly downhill.** §7.6: G falls and E rises simultaneously, so μ is worse everywhere off the symmetric point.

(iii) **Rank reduction: blocked.** §7.5. This is the only place where a *record* was available on paper — $\mu(\zeta_2(5)) \le 7.17740$ from the $M = 5$, $m = 0$ family and $\mu(\zeta_2(7)) \le 12.62404$ from $M = 6$, $m = 1$ — and both are closed.

(iv) **The group: $\delta_{\mathfrak{G}} \le 0$ on the p-adic locus.** Proposition 3.8 gives $\delta_{\mathfrak{G}} = 0$ at the totally symmetric point [PROVED]. The scan adds the empirical half: on the whole p-adic locus the *measured* denominator never falls below its symmetric value, so $\delta_{\mathfrak{G}} \le 0$ throughout [VERIFIED: the cone of §7.6]. The disjoint-support wall of §9.3 is thereby confirmed at $p = 2$ as well as at $p \ge 5$.

Re-verified at 50 decimal digits from (p, r, A, m, ϵ) alone:

$$\begin{array}{llll} \zeta_2(3): & G = 4.158883083359672, & \mathrm{margin} = 1.158883083359672, & \mu = 7.1773988991241796616, \\ \zeta_3(3): & G = 3.295836866004329, & \mathrm{margin} = 0.2958368660043291, & \mu = 22.28144795149432156, \\ \zeta_2(5): & G = 5.545177444479562, & \mathrm{margin} = 0.5451774444795625, & \mu = 20.34265173891448181. \end{array}$$

7.4. **The parity law, and why size never binds at $p = 2$.**

Proposition 7.3 (parity law)**.** *[PROVED], and [VERIFIED: $M = 3, \dots, 7$, exact] For the very-well-poised family (7.1) with any integer insets $e, f \ge 0$ one has $R(-h_0 - t) = -(-1)^M R(t)$,* *hence* $r_{i,h_0-k} = -(-1)^{M+i} r_{i,k}$ *and therefore*

$$\rho_i = 0 \qquad \textit{whenever } i+M \textit{ is even.}$$

Corollary 7.4. *[DERIVED] Combining Proposition 7.3 with* $\rho_1 = 0$ *(Lemma 2.2) and* $\zeta_2(\mathit{even}) = 0$ *(K2), the entire ledger of the* $p=2$ *very-well-poised family collapses to two numbers:*

$$\mathrm{margin}(M,m) = 2M\log 2 - (M+m), \qquad \textit{weights } \begin{cases} \{m+4, m+6, \dots, m+M\}, & M \textit{ even},\\ \{m+3, m+5, \dots, m+M\}, & M \textit{ odd}, \end{cases} \tag{7.2}$$

with rank *equal to the number of weights listed.*

Since $\mathrm{margin}(M,m) = M(2\log 2 - 1) - m \to +\infty$ as $M \to \infty$, we obtain:

Finding 7.5 (size is not the problem at $p=2$; rank is)**.** [COMP: $\mathcal{C}$ at $p=2$] There is a positive margin at *every* weight at $p=2$:

M	m	weights	rank	G	E	margin	μ
3	0	$[3]$	1	4.1589	3	$+1.1589$	7.17740 (Beukers, $\zeta_2(3)$)
4	1	$[5]$	1	5.5452	5	$+0.5452$	20.34265 (LSZ, $\zeta_2(5)$)
4	3	$[7]$	1	5.5452	11	-5.4548	— (the rank-1 $\zeta_2(7)$ slice)
6	1	$[5,7]$	2	8.3178	7	$+1.3178$	12.62404 ("one of $\zeta_2(5), \zeta_2(7)$")
7	0	$[3,5,7]$	3	9.7041	7	$+2.7041$	7.17740
8	1	$[5,7,9]$	3	11.0904	9	$+2.0904$	10.61098
9	0	$[3,5,7,9]$	4	12.4766	9	$+3.4766$	7.17740

The entire obstruction at $p=2$ *is that the form carries* $\lfloor (M-1)/2 \rfloor$ *zeta values.* If the $\zeta_2(5)$-coefficient of the $M=6$, $m=1$ family could be removed, $\zeta_2(7)$ would be irrational with $\mu \le 12.62404$.

Corollary 7.4 also produces the first of the two $\zeta_2(7)$ walls. Rank 1 forces $M \in \{2,3\}$ (m even) or $M \in \{3,4\}$ (m odd), hence $M \le 4$; then weight 7 needs $m=3$ ($M=4$) or $m=4$ ($M=3$), and in both cases $m \ge A-1$, so rule (R2) of Finding 5.5 bites and the second prime band appears.

> **Wall #1.** Every rank-1 route to weight 7 at $p=2$ must buy the weight with derivatives, and every such derivative order switches the second denominator band on. This is structural, not numerical.

7.5. **The rank wall.** Removing an unwanted coefficient from a rank-R form requires an operator

$$L = \sum_{j,l,d} c_{j,l,d}\, n^d\, (n \to n+l)\ (\text{inset-form } j) \tag{7.3}$$

with *polynomial* coefficients. Anything of size e^{Gn} doubles β and gives $\alpha - 2\beta = -2E < 0$; this is the classical elimination tax, and it is not a modelling assumption: combining the $M=6$ form with the LSZ form explicitly gives margin $11.09 - 25.86 = -14.78$. So the question is precisely whether $\ker(\rho_{w'}) \subseteq \ker(\rho_w)$ inside the difference module generated by the family.

Finding 7.6 (five elimination searches, 0/300)**.** [EXCLUDED: exact linear algebra over $\mathbb{F}_q$, $q = 2^{61}-1$; zero tolerance; degree and shift bounds as tabulated; every hit re-verified over $\mathbb{Q}$]

test	family	kill $\to$ keep	forms	L	D	unknowns	dim ker	keep $\neq 0$	control
shifts only	M=6, m=1	$\rho_3 \to \rho_5$	1	4	10	55	0	—	—
shifts, wider	M=6, m=1	$\rho_3 \to \rho_5$	1	6	12	91	0	—	—
inset cone	M=6, m=1	$\rho_3 \to \rho_5$	4	3	8	144	66	**0**	66/66
inset cone, wider	M=6, m=1	$\rho_3 \to \rho_5$	6	4	6	210	120	**0**	120/120
the $\zeta_2(5)$ record route	M=5, m=0	$\rho_2 \to \rho_4$	4	3	8	144	77	**0**	77/77
the rank-3 route	M=7, m=0	$\rho_2, \rho_4 \to \rho_6$	4	3	8	144	36	**0**	36/36
the rank-3 route	M=8, m=1	$\rho_3, \rho_7 \to \rho_5$	4	3	6	112	1	**0**	1/1

Totalling the five searches with a non-trivial kernel: $66 + 120 + 77 + 36 + 1 = 300$ kernel vectors, of which **0** keep the wanted coefficient, against positive controls at **300/300**.

Remark 7.7 (the positive control)**.** The control evaluates the *same* kernel on a coefficient sequence taken from a *different* family ($M = 8$, resp. $M = 6$). Every kernel vector gives a nonzero value there. So the machinery does detect survivors when survivors exist, and the negative result is about the mathematics and not about the search. The two "shifts only" rows have $\dim\ker = 0$ and hence carry no control; they are reported for completeness.

Wall #2. Every polynomial-coefficient operator that annihilates the $\zeta_2(5)$ coefficient annihilates the $\zeta_2(7)$ coefficient as well — and the same in all four families tested, including the $\zeta_2(5)$-record route. The coefficients generate the same difference module; these forms are irreducible.

The statement that would upgrade Wall #2 to a theorem is Open Lemma 9.2 below.

7.6. **The length cone at $p = 2$: the tension, measured.** The remaining freedom in $\mathcal{C}$ is the length cone — $e_j = \eta_j n$, $f_j = \phi_j n$, the Rhin–Viola direction. With $e, f \geq 0$ forced,

$$\sum \lambda = M - 2\sum \eta \leq M, \qquad \sum \nu = M - 2\sum \phi \leq M, \qquad G = \left(\sum \lambda + \sum \nu\right) \log 2 \leq 2M \log 2,$$

with equality *exactly* at the symmetric point — where $C_1 = 0$ and $\delta_{\mathfrak{G}} = 0$.

Finding 7.8. [VERIFIED: $M = 6$, $m = 1$; C_1 de-biased against the symmetric control at the same h_0; E measured] Along the length cone:

profile $(\eta; \phi)$	$\sum \lambda$	$\sum \nu$	G	C_1	E	margin
symmetric	6.000	6.000	8.3178	+0.0000	7	+1.3178
one numerator brick shortened	5.667	6.000	8.0867	−0.6408	9	−0.9133
two numerator bricks shortened	5.333	6.000	7.8557	−1.1874	10	−2.1443
one numerator + one pole	5.667	5.667	7.8557	−0.0260	7	+0.8557
stronger asymmetry	5.000	5.667	7.3936	−1.3089	9	−1.6064

G falls *and* E rises. The symmetric point is strictly optimal.

Because E is measured here, this already includes whatever Φ_n saving is actually present. *At $p = 2$ the $C_1/\delta_{\mathfrak{G}}$ tension is not a near miss: $\delta_{\mathfrak{G}}$ is 0 and the cone is strictly downhill.*

7.7. $\zeta_2(7)$**: the verdict.**

Finding 7.9. [COMP: $\mathcal{C}$] No configuration in $\mathcal{C}$ pushes the $\zeta_2(7)$ margin positive. The best margin is -5.4548, at $p = 2$, $r = 1$, $A = 4$, $m = 3$, $\epsilon = 1$, all shifts $\frac{1}{2}$:

$$G = 5.5451774444795624753 = 8\log 2, \quad \alpha = 11.090354888959124951 = 16\log 2 = 2G,$$

$$\beta = G + 11 = 16.545177444479562475, \quad \text{margin} = -5.4548225555204375247.$$

The uncorrected ledger would report $E_{\text{led}} = 7$ and margin $= -1.4548$; the four-unit gap is the second prime band of Finding 5.5, which no shape in the cone removes (Finding 5.7). Re-verified at 50 digits, and E_{true} re-verified at larger n: the bands are $(7, 4)$ at $n = 48, 60, 72$, and the measured $\log(\text{denom})/n$ is $9.72, 10.18, 10.21$, rising to 11 from below.

7.8. **The escape hatch, and why it is shut.** By Proposition 6.1 the $p \geq 5$ deficit is $-E$ where its hypothesis holds, and at worst $A\,p\log p/(p-1)^2 - E$ in general (Corollary 6.4); either way the single inequality in the derivation is $\alpha = \min_j v_p(S_n(j/p^r))$: a twisted sum could in principle be *more* p-adically small than its worst summand, through systematic cancellation between cosets. Nothing in the literature does this and the ledger has no term for it. We tested it.

Since the partial fractions $r_{i,k}$ do not depend on the integration shift, one exact computation serves every coset; the J_u come from the exact truncated Volkenborn series. Two questions were asked.

(1) The sums the theory actually needs. The plain sum and all $\varphi(p^r)$ Teichmüller twists $\sum_j \omega(j)^k S_n(j/p^r)$:

configuration	$\min_j v_p$ at the largest n	best twisted sum	gain
p=5, A=2, bricks on 1/5, n=27	70	71	+1
p=5, A=4, bricks on 1/5, n=21	109	109	0
p=5, A=4, spread over 4 cosets, n=21	134	136	+2
p=5, A=4, 2+2 reflection split, n=21	109	110	+1
p=5, A=4, m=1, ϵ=1, n=21	109	110	+1
p=7, A=3, ϵ=1, on 1/7, n=18	65	65	0
p=7, A=3, spread, n=18	65	66	+1
p=5, r=2 (20 cosets), n=30	139	140	+1

Gains are 0 to 3 in *absolute* terms and do not grow with n. The requirement is $\Omega(n)$; specifically $E \cdot n \geq 3n$.

(2) The sharp question. Normalise $u_n := (S_n(j))_j/p^{\min_j v_p}$, so that some coordinate is a unit. A *fixed* $c \in \mathbb{Z}_p^{\varphi}$ gains $\Omega(n)$ for all n if and only if all the directions u_n lie in one hyperplane to depth $\Omega(n)$; that depth is $H := \max_i d_i$ of the Smith normal form of the matrix (u_n) over $\mathbb{Z}_p$ (unit-tested on synthetic data). Measured, once the number of n-values exceeds the number of cosets:

configuration	H
$p = 5,\ A = 2$, on 1/5	$14, 14, 14, 14$ at $n = 15, 18, 21, 24, 27$ (while $\min v_p$ grows $40 \to 70$)
$p = 5,\ A = 4$, on 1/5	$26, 26, 26$ at $n = 15, 18, 21$
$p = 5,\ A = 4$, spread	$0, 0, 0$ at $n = 15, 18, 21$
$p = 5,\ A = 4$, 2+2	$20, 20, 20$ at $n = 15, 18, 21$
$p = 5,\ A = 4,\ m = 1,\ \epsilon = 1$	$28, 28, 28$ at $n = 15, 18, 21$
$p = 7,\ A = 3$, on 1/7	$11, 11, 11$ at $n = 14, 16, 18$
$p = 7,\ A = 3$, spread	$10, 10, 10$ at $n = 14, 16, 18$

H is constant in n while $\min_j v_p(S_n)$ grows linearly. Hence no fixed $\mathbb{Z}_p$-combination of the cosets — twisted, plain or arbitrary — gains more than a bounded number of digits.

Finding 7.10 (the hatch is shut)**.** [EXCLUDED: exact p-adic arithmetic; tolerance zero (exact valuations); range: $p \in \{5, 7\}$, $r \in \{1, 2\}$, $A \in \{2, 3, 4\}$, $m \in \{0, 1\}$, $\epsilon \in \{0, 1\}$, bricks aligned / spread / reflection-split, $n \leq 27$ at $p = 5$, $n \leq 18$ at $p = 7$, $n \leq 30$ at $p = 5$, $r = 2$] The gain over the per-coset minimum is ≤ 3 absolutely, and the hyperplane depth is ≤ 28 and constant in n, while the requirement grows like $E \cdot n$.

Remark 7.11 (what this is and is not)**.** This is not a proof of impossibility. It is a fair sweep with a clean negative and a stated range. One caveat is recorded in place: the $p = 5$, $r = 2$ run has 20 cosets and only 9 values of n, so its H is rank-deficient and uninformative; its ω-twist gain is +1, which is the informative half of that row.

7.9. **Non-vanishing, for the record.** No winner exists in $\mathcal{C}$, so nothing here needs a non-vanishing argument. For the record, the two objects that would have needed one:

- the $M = 6$, $m = 1$ rank-2 family (margin $+1.3178$) inherits LSZ's route: the parity law gives $\rho_2 = \rho_4 = \rho_6 = 0$ identically, and the surviving ρ_3, ρ_5 are nonzero at every n tested [VERIFIED: exact, $n \leq 60$]; a Casoratian argument in the style of [14, Lemma 15(b)] would be the natural proof. It is moot, by Finding 7.6;
- the $m = -1$ route is dead on smallness, not on vanishing — though we note the degeneracy $\rho_0 \equiv 0$ identically for M odd with $m = -1$, and for M even with m even.

Orbit moves preserve non-vanishing (Proposition 3.15); that remains available and, in the present cone, unused.

8. The margin map

We now assemble the whole picture. The map is presented three times: at rank 1 with the ledger's E (which is an independent re-derivation, and reproduces the earlier table exactly); at rank 1 with the corrected, measured E_{true}; and along the rank axis, which is where the dichotomy becomes visible.

8.1. Rank 1, ledger E.

Finding 8.1. [COMP: $r \leq 3$, $A \leq 14$, $0 \leq m \leq w + 2$, $\epsilon \in \{0, 1\}$, all coset spreads, $p \leq 13$, $w \leq 9$] Best rank-1 margin per cell, with $E = A + m + 1 - \epsilon$:

$p \setminus w$	3	5	7	9
2	$+$**1.1589** (known)	$+$**0.5452** (known)	-1.4548	-3.4548
3	$+$**0.2958** (known)	-1.7042	-3.7042	-5.7042
5	-3.0000	-2.9882	-4.9882	-6.9882
7	-3.0000	-4.0000	-6.0000	-8.0000
11	-3.0000	-4.0000	-6.0000	-8.0000
13	-3.0000	-4.0000	-6.0000	-8.0000

The three known results come out positive, and *only* those three.

Remark 8.2 (on reading this table)**.** The optimised quantity is the margin, not μ. A cell may attain its best margin at a configuration with a larger α, hence a worse μ, than the published one; the μ-values quoted in Theorem 3 and in Finding 8.4 are those of the published configurations, which are separately optimal for μ over $\mathcal{C}$ (§7.3).

Remark 8.3 (a fitted law that misled)**.** An earlier working extrapolation of the ledger, margin $= (w + 3) \log p/(p - 1) - w$, matches all three anchors and predicts $\zeta_5(3)$ at -0.586 and $\zeta_3(5)$ at -0.606. It is an artefact. The anchors sit at $(p, r, A) = (2, 2, 2)$, $(2, 1, 4)$ and $(3, 1, 2)$, and $A(r + \frac{1}{p-1})$ happens to equal $(w + 3)/(p - 1)$ at exactly those points. The fit has no r, no A, and — fatally — no coset count. The first-principles values are -3.000 and -1.704. We record this because the 0.586 figure drove a subsequent modelling step, and its refutation is the content of §9.3.

8.2. Rank 1, corrected E_{true}.

Finding 8.4 (the honest map)**.** [COMP: as Finding 8.1, with E measured per Finding 5.5]

p	w	ledger margin	E_{led}	E_{true}	**true** margin	μ	configuration	
2	3	+1.1589	3	3	**+1.1589**	7.17740	r=1, A=3, m=0, ϵ=1	**[published]**
2	5	+0.5452	5	5	**+0.5452**	20.34265	r=1, A=4, m=1, ϵ=1	**[published]**
2	7	−1.4548	7	**11**	**−5.4548**	—	r=1, A=4, m=3, ϵ=1	
2	9	−3.4548	9	15	−9.4548	—	r=1, A=4, m=5, ϵ=1	
3	3	+0.2958	3	3	**+0.2958**	22.28145	r=1, A=2, m=0, ϵ=0	**[published]**
3	5	−1.7042	5	6	**−3.8028**	—	r=3, A=4, m=1, ϵ=0	
3	7	−3.7042	7	12	−8.7042	—	r=1, A=2, m=4, ϵ=0	
3	9	−5.7042	9	16	−12.7042	—	r=1, A=2, m=6, ϵ=0	
5	3	−3.0000	3	3	**−3.0000**	—	r=1, A=2, m=0, ϵ=0	
5	5	−2.9882	5	6	−3.9882	—	r=1, A=4, m=1, ϵ=0	
5	7	−4.9882	7	12	−9.9882	—	r=1, A=4, m=3, ϵ=0	
5	9	−6.9882	9	16	−13.9882	—	r=1, A=4, m=5, ϵ=0	
7	3	−3.0000	3	3	**−3.0000**	—	r=1, A=2, m=0, ϵ=0	
7	5	−4.0000	4	5	−5.0000	—	r=1, A=3, m=1, ϵ=0	
7	7	−6.0000	6	11	−11.0000	—	r=1, A=3, m=3, ϵ=0	
7	9	−8.0000	8	15	−15.0000	—	r=1, A=3, m=5, ϵ=0	
11, 13	3/5/7/9	−3/−4/−6/−8		3/5/11/15	−3/−5/−11/−15	—	as $p=7$	

The headline row. The three published results are exactly the three positive cells, and they are untouched by every correction. After the correction the nearest miss in the whole family is no longer $\zeta_2(7)$: it is $\zeta_5(3)$ and $\zeta_7(3)$ at exactly -3.000 — and those two are the *hardest* cells structurally (Proposition 6.1, whose hypothesis holds at both, since $A = 2 < \varphi(p)$ there), because there the deficit is $-E$ exactly, with the p-adic smallness cancelling the archimedean growth to the last digit.

Remark 8.5 (the one inherited assumption)**.** Rule (R2) of Finding 5.5 was verified for the single-shift shape; the multi-coset spreads at $p \geq 5$ inherit it by assumption, not by measurement. The $p \geq 5$ conclusions do not depend on it: those cells are 3 to 15 short either way. It matters only for the E_{true} column at $p \geq 5$, and it is Open Lemma 9.1.

8.3. **The rank axis, and the dichotomy.**

Finding 8.6 (best margin at rank $\leq R$)**.** [COMP: as Finding 8.1; ledger E; $R > 1$ gives only "one of R values"]

p	w	rank 1	rank 2	rank 3	rank 4	min rank for margin > 0
2	3	+1.1589	+3.2383	+4.3178	+5.3972	**1**
2	5	+0.5452	+2.2383	+4.3178	+5.3972	**1**
2	7	−1.4548	**+1.3178**	+2.7041	+4.3972	**2**
2	9	−3.4548	−0.6822	**+2.0904**	+3.4766	**3**
3	3	+0.2958	+1.9438	+2.5917	+3.2396	**1**
3	5	−1.7042	**+0.9438**	+2.5917	+3.2396	**2**
3	7	−3.7042	−1.0562	**+0.5917**	+2.2396	**3**
3	9	−5.7042	−3.0562	−1.4083	**+0.2396**	**4**
5	3	−3.0000	−1.9882	−1.9882	−1.9882	none ≤ 4
5	5	−2.9882	−1.9882	−1.9882	−1.9882	none ≤ 4
5	7	−4.9882	−3.9882	−3.9882	−3.9764	none ≤ 4
5	9	−6.9882	−5.9882	−4.9764	−3.9764	none ≤ 4
7	3	−3.0000	−3.0000	−3.0000	−3.0000	none ≤ 4
7	5	−4.0000	−4.0000	−3.7298	−3.7298	none ≤ 4
7	7	−6.0000	−4.7298	−3.7298	−3.7298	none ≤ 4
7	9	−8.0000	−6.7298	−5.7298	−5.7298	none ≤ 4
11, 13	3/5/7/9	$-3/-4/-6/-8$, identical at every rank ≤ 4				none ≤ 4

The dichotomy. Read the table as: *at $p \in \{2,3\}$ the wall is rank; at $p \geq 5$ the wall is size.* At $p \in \{2,3\}$ every cell becomes positive at a large enough rank — the form exists and is small enough, it just carries too many zeta values. At $p \geq 5$ no cell becomes positive at any rank ≤ 4: the form is not small enough, and rank is irrelevant. These are two different obstructions, and the campaign literature has been treating them as one.

8.4. **The structural ceiling.**

Finding 8.7. [DERIVED] + [VERIFIED: exact] The rank-1 ceiling of Corollary 6.4 per prime:

p	$p\log p/(p-1)^2$	verdict at rank 1
2	1.386294	size never binds: margin $= M(2\log 2 - 1) - m \to +\infty$
3	0.823959	margin $\leq A(0.8240 - 1) - 1 < 0$ for every A
5	0.502949	margin $\leq A(0.5029 - 1) - 1 < 0$
7	0.378371	margin $\leq A(0.3784 - 1) - 1 < 0$
11	0.263768	< 0
13	0.231558	< 0

The single number $p\log p/(p-1)^2$ crossing 1 between $p = 2$ and $p = 3$ is the dichotomy in one line.

Note that the ceiling is a *derived* bound, valid for every A, r and m in regime T, and hence covers the whole rank-1 half of the map at $p \geq 3$ without any search. The searches of §7 are needed only for the rank axis, for E_{true}, and for the $p = 2$ cells.

9. Discussion

9.1. **The two open lemmas.** Two of the obstructions above are *fair sweeps with clean negatives*, not theorems. Each rests on a single clean statement which we believe and do not prove. Proving either would upgrade an obstruction to a theorem, and both look like the sort of statement the existing literature is equipped to prove.

Open Lemma 9.1 (rule R2)**.** [OPEN] In the family of Definition 4.1, the exponent of the primes $\ell \in (n, 2n)$ in the denominator of ρ_0 is $c = m + 1$ when $m \geq A - 1$ and $c = 0$ otherwise.

Status: [VERIFIED: $p \in \{2,3,5,7\}$, $r \in \{1,2\}$, $A \leq 5$, $m \leq 4$, $\epsilon \in \{0,1\}$, $n = 32,\ldots,48$; 0 violations; single-shift shape]. *Content:* it is a statement about when the odd primes in $(n, 2n)$ cancel in $\sum_{i,k} r_{i,k}(i)_{m+1}T_{k,i+m+1}$. *Why it matters:* it would make Correction 2 of §5.5 [PROVED] rather than measured; and, independently of this paper, it would sharpen every published d_n-bound in the $m \geq 2$ range. It is a Nesterenko/Zudilin-style denominator lemma in the spirit of [17, Lemma 16] and [13, Lemmas 26–29], and it looks provable by their methods. A second, weaker open point is R2 for multi-coset spreads at $p \geq 5$, which §8.2 assumes rather than measures.

Open Lemma 9.2 (the kernel inclusion)**.** [OPEN] For the very-well-poised difference module generated by the family (7.1) at $p = 2$,

$$\ker \rho_{w'} \subseteq \ker \rho_w \qquad \text{for } w' < w \text{ of the same parity class,}$$

where the kernels are taken inside the module of polynomial-coefficient difference operators (7.3).

Status: [EXCLUDED: five searches, $0/300$ kernel vectors keep the wanted coefficient; positive controls $300/300$; $\mathbb{F}_q$ with $q = 2^{61} - 1$; every hit re-verified over $\mathbb{Q}$] (Finding 7.6). *Why it matters:* it would turn the $\zeta_2(7)$ obstruction into a theorem. *What it is, conceptually:* this is the exact

p-adic analogue of the phenomenon that "one of $\zeta(5), \dots, \zeta(11)$ is irrational" has never been sharpened to a single value — the window is never narrowed. In the archimedean world that failure has never been explained structurally; here it acquires a precise algebraic formulation, namely that the coefficients of the very-well-poised family generate a single irreducible difference module. We regard the availability of that formulation as one of the more useful by-products of the scan.

We note the asymmetry between the two: Open Lemma 9.1 is a computation waiting to be done, while Open Lemma 9.2 is a structural statement whose proof would be of independent interest.

9.2. **The calibrated step, stated and then refuted.** This subsection contains the only [CALIBRATED] statement in the paper, and its refutation. We keep both, because the refutation is the structural finding of §9.3 and it is not visible without the model it refutes.

The model. Optimising $\delta_{\mathfrak{G}}$ alone is the wrong objective: leaving the totally symmetric point to buy $\delta_{\mathfrak{G}}$ *also* costs p-adic smallness. Calibrating the archimedean side against [17, Lemma 12], anchored to Zudilin's printed $C_0 = 47.15472079$, $C_1 = 48.46940964$ at the Rhin–Viola optimum, one obtains

point	$C_0/$ budget	$\delta_{\mathfrak{G}}/$ budget	margin / budget $= (C_0 -$ budget $+\delta_{\mathfrak{G}})/$ budget
symmetric (Apéry)	1.175165	0	+0.175165
Rhin–Viola optimum	0.943094	0.403754	+0.346848
max-$\delta_{\mathfrak{G}}$ point $(4,7,8,11;\dots)$	0.551327	0.412182	−0.036 (construction dies)
max-$\delta_{\mathfrak{G}}$ that still works	0.647086	0.437758	+0.084844
best combined $(25,26,27,28;1,8,48,49)$	1.054454	0.326954	**+0.381408**

[CALIBRATED] **Equal-degradation model.** Assume the p-adic smallness ratio $(\alpha_p -$ growth$)/$ budget degrades off the symmetric point by the same amount as the archimedean $C_0/$ budget does. With the symmetric-point values $C_0/$ budget $= 1.175165$ and $(\alpha_p -$ growth$)/$ budget $= 0.804734$, whose difference is $\Delta = 0.370431$, this gives

$$\frac{p\text{-adic margin}}{\text{budget}} = \frac{C_0 - \text{budget} + \delta_{\mathfrak{G}}}{\text{budget}} - 0.370431,$$

so that $\zeta_5(3)$ would be reached as soon as $\max(C_0 - \text{budget} + \delta_{\mathfrak{G}})/\text{budget} > 0.370431$. Hill-climbing over integral directions (six independent seeds, scales to $\sum \alpha \approx 250$) attains **0.381408** at $\alpha = (25, 26, 27, 28)$, $\beta = (1, 8, 48, 49)$, i.e. a p-adic margin of $+0.010977$ per unit budget, $+0.0329$ absolute at budget 3, against -0.5858 at the symmetric point. On this model the orbit method removes 106% of the deficit.

The crossing is entirely inside the modelling step, and we flagged it as such. It does not survive.

Finding 9.3 (the crossing fails, by a factor of about three)**.** [VERIFIED: exact; three directions; ledger of §4.9] Evaluating $\alpha_p -$growth by (4.10) rather than by the equal-degradation assumption, at $p = 5$, $r = 1$:

direction	budget	$\delta_{\mathfrak{G}}$	C_1	$\alpha_p -$ growth	requirement
crossing $(25,26,27,28;1,8,48,49)$	72	23.5407	77.2577	−28.97	$> +48.46$
Rhin–Viola optimum	50	20.1877	48.4694	−12.26	$> +29.81$
Apéry/Ball symmetric	3	0	3.5255	−1.51	$> +3.00$

At the crossing point the shortfall is 77.43. Even the most generous accounting — an ideal even coset split and C_1 ignored entirely — gives $42.25 < 48.46$, still a failure. The break-even condition $(r + \frac{1}{p-1}) \log p[\sum \nu - \sum_{\text{mis}} \lambda] > \text{budget} - \delta_{\mathfrak{G}} + C_1$ has left side at most $(1 + \frac{1}{4}) \log 5 \cdot \frac{1}{2} \sum \nu =$

$1.006 \sum \nu$ and right side $\approx 3.0 \sum \nu$ at the tested directions — a factor of about 3, not the 16% relative degradation the model allowed.

9.3. **The structural finding: disjoint support.** The equal-degradation model omitted two terms. One is bookkeeping; the other is structural and, we believe, is the real content of this paper's negative half.

(1) C_1 — and this one is structural. By Remark 4.15, Bel's p-adic criterion needs $\alpha >$ growth + denominators, whereas the archimedean Rhin–Viola criterion needs only $C_0 > C_2$: archimedeanly, C_1 enters the *measure* but never the *irrationality condition*. So the p-adic side pays the coefficient growth. And by Proposition 4.9, $C_1 = 0$ holds *exactly* on the degenerate locus where the numerator bricks overlie the poles — which is *exactly* where Proposition 3.8 gives $\delta_{\mathfrak{G}} = 0$.

> $\delta_{\mathfrak{G}} > 0$ **and** $C_1 = 0$ **have disjoint support.** Measured: $C_1/$ budget $= 1.175$ (symmetric), 0.969 (Rhin–Viola optimum), 1.073 (the crossing point) — each of which alone exceeds the entire available $\delta_{\mathfrak{G}}/$ budget ≤ 0.4378 (Finding 3.9). Moreover $C_1 \geq C_0 > C_2$ at every direction measured — with equality $C_1 = C_0 = 4\log(1+\sqrt{2}) = 3.5255$ at the symmetric point, and strictly at the other two — so the p-adic requirement $\alpha_p > C_1 + C_2$ is more than twice the archimedean one (2.17, 2.63 and 2.59 times C_2 at the three directions).

This is a genuinely structural obstruction rather than a numerical one. The group saving $\delta_{\mathfrak{G}}$ is a *denominator* phenomenon and lives on asymmetric directions; the vanishing of the archimedean saddle is a *growth* phenomenon and lives on the symmetric locus. The p-adic criterion needs both at once, and they do not overlap. The same tension appears at $p = 2$ (Finding 7.8), where it is not even a near miss: there $\delta_{\mathfrak{G}}$ is identically 0 on the cone and E rises as G falls.

(2) The coset defect. For ζ_p at $p \geq 5$ a numerator brick aligns with one coset only, so at the worst coset the aligned length is at most $\sum \lambda/c_p$, $c_p = \varphi(q_p)/2$. Measured defect at the three directions of Finding 9.3: 0.50, 0.357, 0.4286 of $\sum \nu$.

Remark 9.4 (what would have had to be true)**.** The equal-degradation model allowed the p-adic smallness ratio to degrade by 16% relative and still win. What the first-principles computation shows is a factor of 3. That is not a near miss that a better direction might close; it is a different regime.

9.4. **What a construction evading the map would need.** The value of a map is that it says where not to walk. Collecting the obstructions, a construction that produced a new p-adic measure would have to violate at least one of the following, and we state each as a design constraint rather than as a prohibition.

(A) **Break the disjoint support of $\delta_{\mathfrak{G}}$ and C_1.** One needs a family in which the archimedean coefficient growth stays at its minimum G while the direction is asymmetric enough to carry a group orbit. Within Definition 4.1 this is impossible, because the vanishing of the saddle *is* co-location (Proposition 4.9). A construction with a different source of coefficient control — e.g. one whose coefficients are integers by construction rather than by clearing denominators, as in cellular-integral constructions [5, 7] — would not be bound by this argument. Brown's Remark 47 [6] asks exactly this question from the determinant side, and our Corollary 2.5 says at least that the group identities are available to any such construction.

(B) **Escape the criterion's growth term.** Alternatively, keep the family and change the criterion: any p-adic irrationality criterion that does not charge for $\max(|a_n|, |b_n|)$, as the archimedean one does not, would immediately change the arithmetic of (4.8). We are not aware of one, and Bel's lemma is the standard tool; but the asymmetry of Remark 4.15 is a property of the criterion, not of the numbers.

(C) **Beat the per-coset minimum by $\Omega(n)$ at $p \geq 5$.** This is the hatch of §7.8. What is needed is a family with $v_p\big(\sum_j S_n(j/p^r)\big) > \min_j v_p(S_n(j/p^r)) + \Omega(n)$. The measured hyperplane depth is bounded and constant, so the cancellation would have to be of a kind that does not correspond to a fixed $\mathbb{Z}_p$-combination — i.e. genuinely n-dependent structure, not a twist.

(D) **Reduce the rank at $p \in \{2, 3\}$.** This is Open Lemma 9.2. What is needed is an elimination with coefficients that are *not* of size e^{Gn} and *not* polynomial-coefficient difference operators within the family — e.g. an elimination that uses forms from a genuinely different family, so that the module argument does not apply. This is what the elimination technique of [9] does archimedeanly, and [13] is the p-adic version; the obstruction there is non-vanishing, not size — which is precisely why it yields "one of" statements and not measures.

(E) **Reach the orbit from a p-adic base point.** Even granting all of the above, the 1920-element orbit is currently attached to integer parameters, and a genuinely p-adic base point reaches it only through the quadratic transformation of Open Lemma 3.14. That lemma is the cheapest missing piece in the entire chain, and it is likely a matter of citing the right classical identity.

9.5. **Where we would look next.** Three suggestions, in increasing order of ambition.

(1) *Prove Open Lemma 3.14* and Open Lemma 9.1. Neither requires new ideas; both would convert measured statements into proved ones, and R2 sharpens published d_n-bounds independently of this paper's subject.

(2) *Attack Open Lemma 9.2*, which is the p-adic shadow of the never-sharpened window. A proof would explain, in a precise algebraic sense, why "one of $\zeta(5), \ldots, \zeta(11)$" resists sharpening; that would be of interest well beyond the p-adic setting.

(3) *Leave the family.* Every wall in §8 is a wall of Definition 4.1. The three constraints that generate them — rank-1 forms come from very-well-poised shapes, coefficient control comes from clearing d_n, and smallness is a minimum over cosets — are all properties of hypergeometric brick constructions. Cellular integrals [5, 7] and the determinant criterion [6] organise coefficient control differently, and the Transfer Principle guarantees that any group identity available to them archimedeanly is available p-adically. That is, we think, the most promising reading of the negative results collected here.

9.6. **Summary of what is proved, measured and open.**

statement	class
Transfer Principle (Thm. 1) and its two corollaries	[PROVED]
$\delta_{\mathfrak{G}}$ is completion-independent (Prop. 2.8)	[DERIVED]
$\delta_{\mathfrak{G}} = 0$ at the totally symmetric point (Prop. 3.8)	[PROVED]
$\lvert\mathfrak{G}\rvert = 1920$; transfer identity at 1321 integer orbit points	[VERIFIED: 0 failures]
transfer identity at 780 half-integer orbit points	[VERIFIED: 0 failures]
$\mu(\zeta(3)) \le 5.51389063$ from the pipeline	[VERIFIED: 8 digits]
LSZ shape class has no internal transfer identity	[EXCLUDED: 2414 points, exact]
the Gauss-duplication bridge (Open Lemma 3.14)	[VERIFIED: 13 instances] / [OPEN]
classification $\varphi(p^r) = 2$ (Thm. 2)	[PROVED]
margin $= -E$ at $p \ge 5$ (Prop. 6.1)	[DERIVED] + [VERIFIED: $p = 5$]
ledger reproduces the three published measures	[VERIFIED: every digit]
$m = -1$ loses the doubling (Prop. 5.4)	[PROVED] + [VERIFIED: $n \le 22$]
the second prime band, rules R1 and R2	[VERIFIED: 0 violations] / [OPEN] (R2)
parity law (Prop. 7.3); $\operatorname{margin}(M, m) = 2M\log 2 - (M+m)$	[PROVED] + [DERIVED]
optimality of the three measures over $\mathcal{C}$ (Thm. 3)	[COMP: $\mathcal{C}$]
the rank wall (Open Lemma 9.2)	[EXCLUDED: 0/300, controls 300/300] / [OPEN]
the coset hatch is shut	[EXCLUDED: range in Finding 7.10]
the margin map and the dichotomy	[COMP: stated domain]
$\delta_{\mathfrak{G}} > 0$ and $C_1 = 0$ have disjoint support	[DERIVED] + [VERIFIED: measured]
the equal-degradation crossing	[CALIBRATED], and refuted

Appendix A. Reproduction record

Every numerical statement in this paper comes from one of the programs listed below. They are plain Python 3 (`fractions`, `itertools`, with `sympy` for factorisation, `numpy`/`scipy` for the digamma quadrature in (3.7), and `mpmath` for the 50-digit re-verifications). All arithmetic reported as exact is exact; floating point appears only in the final evaluation of $\log p$, ψ and the two integrals of (3.7).

A.1. **Programs.**

The group (§3).

program	contents	runtime
`g_group.py`	the group of order 1920 by BFS closure; the function Λ; $\delta_{\mathfrak{G}}$; the m-parameters; the Rhin–Viola validation $\mu(\zeta(3)) \le 5.51389063$; $\delta_{\mathfrak{G}} = 0$ at the symmetric point	0.3 s
`g_verify.py`	exact coefficientwise verification of the transfer identity (Finding 3.4); the Apéry/Ball anchor	4 s
`g_padic.py`	the same test at half-integer parameters (Finding 3.5)	2 s
`g_forms.py`	exact (ρ_0, ρ_i) for the parametrised p-adic family; the LSZ anchors $\rho_{0,3} = 768$, $\rho_{1,0} = -1024$, $\rho_{1,3} = 73728$, $\rho_n = 1, 96, 14944$	< 1 s
`g_search.py`, `g_search2.py`	the negative searches inside the LSZ shape class (Finding 3.12)	2 s, < 1 s
`g_full.py`	exhaustive very-well-poised half-integer proportionality classification; the 13 bridge instances (Finding 3.13)	8 s
`g_opt.py`, `g_cal.py`, `g_hill.py`	maximisation of $\delta_{\mathfrak{G}}/$budget (Finding 3.9); the archimedean calibration and hill-climb of §9.2	89 s, < 1 s, 31 s

The ledger (§4, §5).

program	contents	runtime
`ledger.py`	the ledger (4.12) and the validation gate: all three published measures plus the Lai cross-checks (Finding 5.1)	< 1 s
`family.py`	exact partial fractions, ρ_0 and ρ_i, the Volkenborn J_u, and S_n	(library)
`verify.py`	exact regeneration of the LSZ anchor (Finding 5.2); measured α, growth and E; the $p = 3$ configuration	2 s
`optimize.py`	the exhaustive rank-1 optimiser and the freedom map of §4.11	< 1 s
`freefact.py`	the free-factorial direction; the $p \geq 5$ constant reproduced (Finding 4.17)	< 1 s
`t4_freedom.py`	the alignment law verified numerically at $p = 5$ (Finding 6.3); the regime table; the structural ceiling (Finding 8.7)	< 1 s
`gate_dig2.py`	the evaluation of the calibrated crossing (Finding 9.3)	< 1 s

The scan (§7, §8).

program	contents	runtime
`s_vwp.py`	the $p = 2$ very-well-poised cone with insets; the parity law (Proposition 7.3); the LSZ anchors reproduced	< 1 s
`s_meas.py`	exact measurement of α by truncated Volkenborn series, of the growth, and of E; the $m = -1$ verdict (Finding 5.3)	2 s
`s_den.py`	the true E, prime band by prime band; rules R1 and R2 (Finding 5.5)	1 s
`s_elim.py`	partial fractions modulo $q = 2^{61} - 1$ (validated against exact $\mathbb{Q}$); the inset lattice	(library)
`s_recur.py`, `s_rank.py`	the rank-reduction test and its positive control (Finding 7.6)	see below
`s_cone.py`	the length cone: G, C_1 and E measured off symmetry (Finding 7.8)	3 s
`s_record.py`	the record scan with E and the growth measured (§7.3)	25 s
`s_coset.py`	exact coset valuations, ω-twists, and the Smith-form hyperplane depth (Finding 7.10)	long; see below
`s_map.py`	the margin map, with the ledger E and with E_{true} (§8)	18 s

The rank-reduction searches dominate the total cost. Individual timings, taken from the run logs: the four inset-cone searches of Finding 7.6 took 163 s (144 unknowns), 575 s (210 unknowns), 126 s (144), 198 s (144) and 118 s (112); the underlying ρ-vector construction for the widest case built 668 vectors up to $n = 172$ in 108 s. The coset sweep `s_coset.py` runs eight configurations at $p \in \{5, 7\}$ up to $n = 30$ and is the longest single job; its output is preserved as a log.

A.2. **Reproducing the headline numbers.**

(1) *The three published measures from the tuple alone* (Finding 5.1): run `ledger.py`. Expected output includes `mu <= 7.17739889912418`, `22.28144795149432`, `20.34265173891448`, each tagged `[MATCH]`.

(2) *The Rhin–Viola pipeline validation* (Finding 3.7): run `g_group.py`. Expected: $|\mathfrak{G}| = 1920$; at $(18, 17, 16, 19; 0, 7, 31, 32)$, $\int_0^1 \Lambda \, d\psi = 24.18768530$, $\delta_{\mathfrak{G}} = 20.18768530$, $C_2 = 29.81231470$, $\mu(\zeta(3)) \leq 5.51389063$, orbit of distinct F-multisets $= 118$; and at the symmetric point $\delta_{\mathfrak{G}} = 0$ with orbit size 1.

(3) *The map* (§8): run `s_map.py`. Its five blocks are the rank-1 map, the rank axis, the $p = 2$ detail grid, the structural ceiling, and the corrected rank-1 map; the last reproduces Finding 8.4 row for row.

(4) *The crossing evaluation* (Finding 9.3): run `gate_dig2.py`. Expected: at the crossing point, $\alpha_p - \text{growth} = -28.97$ against a requirement of $+48.46$, shortfall 77.43; and, with C_1 ignored and an ideal split, $42.25 < 48.46$.

(5) *The parity law and the anchors* (Proposition 7.3): run `s_vwp.py`. Expected: zero ρ_i at $i \in \{1, 3\}$, $\{1, 2, 4\}$, $\{1, 3, 5\}$, $\{1, 2, 4, 6\}$, $\{1, 3, 5, 7\}$ for $M = 3, \ldots, 7$, matching the

prediction in each case; and LSZ's printed $\rho_{0,3} = 768$, $\rho_{1,3} = 73728$, $\rho_{0,0} = 0$, $\rho_{1,0} = -1024$ with $\rho_n = 1, 96, 14944$.

A.3. **Four artefacts and one driver bug, recorded.** We record these because a reproducer will meet them, and because suppressing them would misrepresent the evidence.

(a) **The $E : 5 \to 3$ drop at $M = 5$, $n = 24$** is a small-n artefact: at $n = 36, 48, 60$ the exponent is back to 5. Reported in §7.3.

(b) `s_record.py` **prints** `NEW RECORD`**, and this is a finite-n artefact of G, not a record.** The program evaluates $\mu = (G + \text{gain})/\text{ margin}$ with G *measured at* $n = 24$, where $\sum \nu$ has not yet converged to its limit M (e.g. $\sum \nu = 3.125$ instead of 3 at $M = 3$). This inflates G and thereby deflates μ, producing 6.817 against the asymptotic 7.177 for $\zeta_2(3)$ and 17.135 against 20.343 for $\zeta_2(5)$; the comparison label then fires against the published *asymptotic* constants. The mathematically load-bearing line in that output is the one immediately above it, `BEST over the cone`, which reports the *argmin* over the inset cone: it is $e = f = (0, \ldots, 0)$, the symmetric point, in both cases. That is the content of sub-computation (i) of §7.3, and it is what Theorem 3 claims. The asymptotic re-verification at 50 digits, quoted at the end of §7.3, returns the published constants exactly.

(c) `g_verify.py` **prints** `[BROKEN]`**, and this is Finding 3.12, not a failure of Finding 3.4.** Block [C] of that program shifts the *numerator bricks* by θ and re-tests proportionality across the orbit; it reports 10/40 and 5/80 at $\theta = \frac{1}{2}, \frac{1}{5}, \frac{2}{5}, \frac{1}{3}$. That is the expected outcome: shifting the numerator bricks leaves the very-well-poised family for the LSZ shape class, which by Finding 3.12 has no internal transfer identity. The positive half-integer test is a different deformation ($h_4, h_5 \in \mathbb{Z} + \frac{1}{2}$) and is run by `g_padic.py`, which returns 120/120, 180/180, 480/480.

(d) `verify.py`**'s "predicted E" column is wrong for Lai's** A_n (it prints 3 at $s = 0$ and 5 at $s = 1$ against the measured 4 and 7). The measured values are the correct ones, $E = A + m + 1 - \epsilon = 3s + 4$, and they are what §5.3 and the ledger use; only that one display is at fault.

(e) **A driver bug.** `s_record.py` raises a `ValueError` when it formats the comparison line for the two rank-2 control scans, which have no published μ to compare against (the string `"-- (rank 2:  weights {3,5})"` is passed to `float`). The numerical content of the third scan is printed before the exception; the fourth scan does not run. Neither affects any statement in this paper: the rank-2 families are dealt with in §7.5, by `s_rank.py`.

A.4. **What is *not* reproduced by the code.** The proofs (Theorems 1 and 2, Lemma 2.2, and Propositions 2.8, 3.8, 3.10, 3.11, 4.3, 4.6, 3.15, 5.4, 7.3) are mathematical arguments, checked by hand; the code verifies their instances but does not establish them. The two open lemmas of §9.1 are, by construction, not established by any computation here.

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[12] L. Lai, *On the irrationality of certain* 2*-adic zeta values*, Int. J. Number Theory **21** (2025), no. 1, 207–235; arXiv:2304.00816. (The families A_n and B_n are (3.1) and (3.2); the $\triangle$-operator estimate is Lemma 2.4, quoted as Lemma 6 of [14].)

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[17] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), no. 1, 251–291; arXiv:math/0206176. (Lemma 8 is the realisation of the group used in §3.1; Lemmas 7–9 and equation (4.4) give the invariant (3.5); Lemma 12 is the saddle-point constant $C_0 = -f_0(\tau_0)$; Lemma 16 is the denominator lemma; equation (2.4) is the unequal-length form.)